Chapter 1. Multivariate Data and the Geometry of the Sample

Data matrices, descriptive statistics, projections, sphering, Mahalanobis distance, eigendecomposition

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1 Multivariate Data and the Goals of Analysis

Multivariate (multidimensional) data are measurements of two or more variables on each individual or experimental unit, for example a multiple regression data set with one response and several explanatory variables measured on each individual. The goals of analysis fall into two groups:

(i) **Inference about an unknown population**: understand and infer features of the population from which the data were drawn.

(ii) **Prediction of future observations**: predict the value or category of new observations.

Whatever the goal, the first step is to understand the data themselves and the information or patterns hidden in them. Exploratory data analysis (EDA) assesses the quality of the data, finds potential errors and outliers, and reveals the basic structure; visual exploration then gives the insight needed for further analysis and modelling. Numerical summaries are descriptive statistics (mean vector, covariance matrix, correlation matrix); graphical summaries are scatterplot matrices, principal component plots, heatmaps, correlation plots and trellis graphs (scatterplots or line plots drawn separately for subgroups of the data). After EDA we study the probability structure of multivariate data and statistical inference about it.

Example 1.1: Palmer penguins (slides Example 1.1)

Body measurements of penguins from the Palmer archipelago. Observed variables: year (one of 2007, 2008, 2009); species (Adelie, Chinstrap, Gentoo); island (Biscoe, Dream, Torgersen); sex (female, male); and the four numerical variables bill length, bill depth, flipper length and body mass. For n penguins with p numerical and q categorical variables the data form an $n \times (p+q)$ array; here $n = 333$, $p = 4$, and year may be treated as numerical or categorical. Only the numerical variables go into the data matrix $\mathbf{X}$; a categorical variable of interest, such as species, is kept as a separate vector. This is the typical case $n > p$.
A scatterplot of flipper length against body mass, coloured by sex, shows that heavier penguins tend to have longer flippers. Factors that affect overall size (year, species, island) are not shown in this plot.

Example 1.2: Golub gene expression data (slides Example 1.2)

A representative early example (Golub et al., 1999) of using genetic information to predict disease. For 72 leukemia patients the expression levels of 7129 genes (numerical variables) were measured, together with one categorical variable giving the leukemia type (ALL or AML). This is high-dimensional, low-sample-size data ($p = 7129 \gg n = 72$). Scientific question: can the leukemia type be predicted from gene expression alone?
A heatmap shows a numerical data matrix as colours: in the slides' heatmap the 22 columns are selected genes and the 72 rows are patients, and a colour bar on the left marks ALL or AML. A heatmap of all 7129 variables would be far too wide to read.

2 The Data Matrix and Descriptive Statistics

Motivation. Multivariate data are an $n \times p$ matrix, and almost every method in the course is a statement about this matrix. Writing the sample mean and covariance as matrix products lets us use linear algebra instead of indices.

2.1 Data matrix

Each of the $p \geq 2$ numerical measurements on an item is called a **variable**, since its value varies from item to item. Measurements of p variables on n items form an $n \times p$ matrix

$$\mathbf{X} = \begin{pmatrix} x_{11} & \cdots & x_{1p} \\ \vdots & \ddots & \vdots \\ x_{n1} & \cdots & x_{np} \end{pmatrix} = \begin{pmatrix} \mathbf{x}_1^\top \\ \vdots \\ \mathbf{x}_n^\top \end{pmatrix} = [\mathbf{x}_{(1)}, \ldots, \mathbf{x}_{(p)}],$$

where x_{ij} is the value of variable j on item i. The ith row $\mathbf{x}_i = (x_{i1}, \ldots, x_{ip})^\top \in \mathbb{R}^p$ is the **observation vector** of item i (vectors are column vectors, so the row is $\mathbf{x}_i^\top$); the jth column $\mathbf{x}_{(j)} = (x_{1j}, \ldots, x_{nj})^\top \in \mathbb{R}^n$ is the **variable vector** of variable j. Observations are written as vectors so that multivariate statistics can be computed with matrix algebra, and all variables in $\mathbf{X}$ are taken to be numerical; categorical variables are handled separately.

2.2 Sample mean, covariance and correlation

For univariate data the distribution is summarized by the mean, median, variance, standard deviation, range, skewness and kurtosis. For multivariate data we likewise compute the mean and variance of each variable; since the variables are related, the covariance and correlation of each pair are also key summaries. The covariance measures the direction and size of a linear relation, and the correlation is the standardized covariance, taking values in $[-1, 1]$.

Definition 1.3: descriptive statistics

$$\bar{\mathbf{x}} = \begin{pmatrix} \bar{x}_1 \\ \vdots \\ \bar{x}_p \end{pmatrix}, \quad \bar{x}_j = \frac{1}{n} \sum_{i=1}^{n} x_{ij},$$

$$\mathbf{S} = (s_{jk}), \quad s_{jk} = \frac{1}{n-1} \sum_{i=1}^{n} (x_{ij} - \bar{x}_j)(x_{ik} - \bar{x}_k),$$

$$\mathbf{R} = (r_{jk}), \quad r_{jk} = \frac{s_{jk}}{\sqrt{s_{jj}}\sqrt{s_{kk}}}, \qquad \mathbf{R} = \mathbf{D}^{-1/2}\mathbf{S}\mathbf{D}^{-1/2},$$

with $\mathbf{D} = \operatorname{diag}(s_{11}, \ldots, s_{pp}) = \operatorname{diag}(s_1^2, \ldots, s_p^2)$ and $\mathbf{D}^{-1/2} = (\mathbf{D}^{1/2})^{-1} = \operatorname{diag}(s_1^{-1}, \ldots, s_p^{-1})$. $\mathbf{R}$ is defined only when all sample variances s_j^2 are positive. The textbook also uses $\mathbf{S}_n = \frac{n-1}{n}\mathbf{S}$ (divisor n).

Theorem 1.4: properties of the sample covariance

1. $\mathbf{S}$ is symmetric: $s_{jk} = s_{kj}$ for all $j, k \in \{1, \ldots, p\}$.
2. $\mathbf{S} = \dfrac{1}{n-1} \displaystyle\sum_{i=1}^{n} (\mathbf{x}_i - \bar{\mathbf{x}})(\mathbf{x}_i - \bar{\mathbf{x}})^\top$.
3. With the centered data matrix $\mathbf{X}_C = (x_{ij} - \bar{x}_j)_{ij}$, $\mathbf{S} = \frac{1}{n-1}\mathbf{X}_C^\top \mathbf{X}_C$.

Proof. (1) The defining sum is symmetric in j and k.

(2) The (j, k) entry of $(\mathbf{x}_i - \bar{\mathbf{x}})(\mathbf{x}_i - \bar{\mathbf{x}})^\top$ is $(x_{ij} - \bar{x}_j)(x_{ik} - \bar{x}_k)$.

(3) The ith row of $\mathbf{X}_C$ is $(\mathbf{x}_i - \bar{\mathbf{x}})^\top$, and $\mathbf{X}_C^\top \mathbf{X}_C = \sum_i (\mathbf{x}_i - \bar{\mathbf{x}})(\mathbf{x}_i - \bar{\mathbf{x}})^\top$. ■

Remark 1.5: a Gaussian with covariance S always exists

$\mathbf{S}$ is symmetric and positive semidefinite, so a symmetric square root $\mathbf{S}^{1/2}$ exists (§5). With $\mathbf{Z} \sim N_p(\mathbf{0}, \mathbf{I}_p)$ and $\mathbf{X} = \boldsymbol{\mu} + \mathbf{S}^{1/2}\mathbf{Z}$, $\mathrm{E}\mathbf{X} = \boldsymbol{\mu}$ and $\mathrm{Cov}\mathbf{X} = \mathbf{S}^{1/2}\mathbf{I}_p\mathbf{S}^{1/2} = \mathbf{S}$. So there is always a Gaussian distribution with mean $\boldsymbol{\mu}$ and covariance matrix $\mathbf{S}$.

2.3 The centering matrix

Let $\mathbf{1}_n = (1, \ldots, 1)^\top$ be the vector of n ones, $\mathbf{0}_n$ the zero vector, $\mathbf{I}_n$ the $n \times n$ identity, $\mathbf{J}_n = \mathbf{1}_n\mathbf{1}_n^\top$ the $n \times n$ matrix of ones, and $\mathbf{H}_n = \mathbf{I}_n - \frac{1}{n}\mathbf{J}_n$.

Theorem 1.6: matrix forms of $\bar{\mathrm{x}}$ and S (slides Exercise 1.1)

$$\bar{\mathbf{x}} = \tfrac{1}{n}\mathbf{X}^\top\mathbf{1}_n, \qquad \mathbf{X}_C = \mathbf{X} - \mathbf{1}_n\bar{\mathbf{x}}^\top = \mathbf{X} - \tfrac{1}{n}\mathbf{1}_n\mathbf{1}_n^\top\mathbf{X} = \mathbf{H}_n\mathbf{X},$$
$$\mathbf{S} = \tfrac{1}{n-1}\mathbf{X}^\top\mathbf{H}_n\mathbf{X} = \tfrac{1}{n-1}\mathbf{X}_C^\top\mathbf{X}_C,$$

and $\mathbf{H}_n^\top = \mathbf{H}_n$, $\mathbf{H}_n^2 = \mathbf{H}_n$.

Proof. The jth entry of $\frac{1}{n}\mathbf{X}^\top\mathbf{1}_n$ is $\frac{1}{n}\sum_i x_{ij} = \bar{x}_j$, giving the first identity. Hence $\mathbf{X}_C = \mathbf{X} - \mathbf{1}_n\bar{\mathbf{x}}^\top = \mathbf{X} - \frac{1}{n}\mathbf{1}_n\mathbf{1}_n^\top\mathbf{X} = \mathbf{H}_n\mathbf{X}$. Symmetry is clear, and idempotency follows from $\mathbf{1}_n^\top\mathbf{1}_n = n$:

$$\left(\mathbf{I}_n - \tfrac{1}{n}\mathbf{J}_n\right)^2 = \mathbf{I}_n - \tfrac{2}{n}\mathbf{J}_n + \tfrac{1}{n^2}\mathbf{1}_n(\mathbf{1}_n^\top\mathbf{1}_n)\mathbf{1}_n^\top = \mathbf{I}_n - \tfrac{1}{n}\mathbf{J}_n.$$

Therefore $\mathbf{X}_C^\top\mathbf{X}_C = \mathbf{X}^\top\mathbf{H}_n^\top\mathbf{H}_n\mathbf{X} = \mathbf{X}^\top\mathbf{H}_n\mathbf{X}$, and $\mathbf{X}_C^\top\mathbf{X}_C = \sum_i(\mathbf{x}_i - \bar{\mathbf{x}})(\mathbf{x}_i - \bar{\mathbf{x}})^\top = (n-1)\mathbf{S}$. ■

[Supplement] A symmetric idempotent matrix has eigenvalues 0 or 1, so $\mathrm{rank}(\mathbf{H}_n) = \mathrm{tr}(\mathbf{H}_n) = n - \frac{1}{n}\mathrm{tr}(\mathbf{J}_n) = n - 1$: $\mathbf{H}_n$ is the orthogonal projection onto the $(n-1)$-dimensional subspace $\{\mathbf{v} \in \mathbb{R}^n : \mathbf{1}_n^\top\mathbf{v} = 0\}$.

3 Sample Geometry in p-Space

3.1 The data ellipse

Each $\mathbf{x}_i$ is a point in Euclidean space $\mathbb{R}^p$, and the points can be shown in a scatterplot. For bivariate data ($p = 2$) the scatterplot shows the relation between the two variables and their distribution; in the slides' penguin plot of bill length against bill depth, the three species form visibly separate clouds, with marginal histograms on the sides.

The mean vector is the center of gravity of the points (in the penguin plot the group mean vectors are marked by triangles) and the covariance matrix describes their spread. The **data ellipse**

$$\{\mathbf{x} \in \mathbb{R}^p : (\mathbf{x} - \bar{\mathbf{x}})^\top\mathbf{S}^{-1}(\mathbf{x} - \bar{\mathbf{x}}) = c\}$$

for a fixed constant $c > 0$ summarizes the location, shape and size of the distribution; for $p \geq 3$ it is an ellipsoid. Its size reflects the spread and its shape the correlations. The penguin plot uses $c = 5.991 = \chi^2_2(0.05)$, one ellipse per species. If the cloud is not elliptical, the mean vector and covariance matrix alone describe it poorly.

Example 1.7: data ellipses

With mean vector $\mathbf{0}$ and covariance matrices $\mathbf{S}_1 = \begin{pmatrix} 9 & 0 \\ 0 & 1 \end{pmatrix}$, $\mathbf{S}_2 = \begin{pmatrix} 5 & 4 \\ 4 & 5 \end{pmatrix}$, draw the data ellipses and check that they reflect the sizes of the variances and covariances.
[Supplement] For $\mathbf{S}_1$ the axes of the ellipse are the coordinate axes with half-length ratio $3:1$. $\mathbf{S}_2$ has eigenvalues $9, 1$ with eigenvectors $(1,1)^\top/\sqrt{2}$, $(1,-1)^\top/\sqrt{2}$, so its ellipse is the same ellipse rotated by $45°$: the positive covariance tilts it along the line $x_2 = x_1$. In both cases the half-axis lengths are $\sqrt{c\lambda_j}$ (§5).

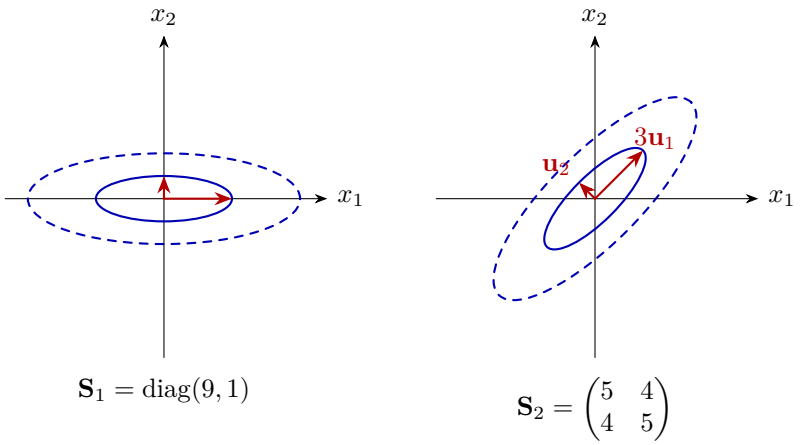


Figure. *Data ellipses* $\mathbf{x}^\top\mathbf{S}^{-1}\mathbf{x} = c$ *for* $c = 1$ *(solid) and* $c = 4$ *(dashed). Both matrices have eigenvalues* 9 *and* 1*, so the ellipses have the same shape; the covariance* 4 *in* $\mathbf{S}_2$ *rotates the axes to* $\mathbf{u}_1 = (1,1)^\top/\sqrt{2}$ *and* $\mathbf{u}_2 = (-1,1)^\top/\sqrt{2}$.

3.2 Subspaces, orthogonal projection and projection scores

The observations $\mathbf{x}_i$ are vectors of the Euclidean space $\mathbb{R}^p$, so transformations and groupings of the observations, and the computation of statistics, can be done with vector operations.

Definition 1.8: subspace, line, basis, affine subspace

- **Subspace**: a line or plane through the origin. For a vector $\mathbf{v}$, the line $\{c\mathbf{v} : c \in \mathbb{R}\}$ is a 1-dimensional subspace of $\mathbb{R}^p$; for linearly independent $\mathbf{v}_1, \ldots, \mathbf{v}_q$, $\mathcal{L}_q = \mathrm{span}\{\mathbf{v}_1, \ldots, \mathbf{v}_q\} = \{\mathbf{V}\mathbf{c} : \mathbf{c} \in \mathbb{R}^q\}$, $\mathbf{V} = [\mathbf{v}_1, \ldots, \mathbf{v}_q]$, is a q-dimensional subspace, said to be spanned by $\mathbf{v}_1, \ldots, \mathbf{v}_q$.
- **Line**: an infinitely extended straight line, as opposed to a line segment, which is a part of it. A flat, infinitely extended surface is a **plane** ($q = 2$) and, for $q = p - 1$, a **hyperplane**.
- **Basis**: the basis $\mathbf{v}_1, \ldots, \mathbf{v}_q$ spanning $\mathcal{L}_q$ is not unique. A basis of mutually orthogonal unit vectors is an **orthonormal basis**.
- **Affine subspace**: lines and planes in Euclidean space, not necessarily through the origin, e.g. the line $\{\mathbf{x} + t\mathbf{v} : t \in \mathbb{R}\}$ through $\mathbf{x}$ in the direction $\mathbf{v}$, or the plane $\{\mathbf{x} + c_1\mathbf{v}_1 + c_2\mathbf{v}_2 : c_1, c_2 \in \mathbb{R}\}$. The affine subspace $\mathcal{L}_q(\mathbf{x}) = \mathbf{x} + \mathcal{L}_q$ is the subspace $\mathcal{L}_q$ translated by $\mathbf{x}$. It is not a vector space in general (it need not contain the origin) but has the same shape as $\mathcal{L}_q$; for every affine subspace there is a parallel subspace of the same dimension. Its representation by $(\mathbf{x}; \mathbf{v}_1, \ldots, \mathbf{v}_q)$ is not unique: affine subspaces generated by different $(\mathbf{x}; \mathbf{v}_1, \ldots, \mathbf{v}_q)$ and $(\mathbf{x}'; \mathbf{v}'_1, \ldots, \mathbf{v}'_q)$ may coincide.

Definition 1.9: orthogonal projection and projection score

The orthogonal projection of $\mathbf{x}$ onto the line $\text{span}\{\mathbf{v}\}$, and the projection score, are

$$P_{\mathbf{v}}(\mathbf{x}) = \frac{\mathbf{v}\mathbf{v}^\top}{\mathbf{v}^\top\mathbf{v}}\mathbf{x} = s_{\mathbf{v}}(\mathbf{x})\frac{\mathbf{v}}{\sqrt{\mathbf{v}^\top\mathbf{v}}}, \qquad s_{\mathbf{v}}(\mathbf{x}) = \frac{\mathbf{v}^\top\mathbf{x}}{\sqrt{\mathbf{v}^\top\mathbf{v}}},$$

where $\mathbf{v}/\sqrt{\mathbf{v}^\top\mathbf{v}}$ is the unit vector in the direction of $\mathbf{v}$. The $p \times p$ matrix $\mathbf{P}_{\mathbf{v}} = \mathbf{v}\mathbf{v}^\top/(\mathbf{v}^\top\mathbf{v})$ is the **projection matrix**: multiplying any $\mathbf{x}$ by it gives the projection onto $\text{span}\{\mathbf{v}\}$. The score $s_{\mathbf{v}}(\mathbf{x})$ is the distance from the origin to $P_{\mathbf{v}}(\mathbf{x}) \in \text{span}\{\mathbf{v}\}$, with sign: positive in the direction of $\mathbf{v}$, negative in the opposite direction (a signed distance).
The projection onto the subspace $\mathcal{L}_q = \text{span}\{\mathbf{v}_1, \ldots, \mathbf{v}_q\}$ is $P_{\mathcal{L}_q}(\mathbf{x}) = \mathbf{P}_{\mathbf{V}}\mathbf{x}$ with $\mathbf{P}_{\mathbf{V}} = \mathbf{V}(\mathbf{V}^\top\mathbf{V})^{-1}\mathbf{V}^\top$, $\mathbf{V} = [\mathbf{v}_1, \ldots, \mathbf{v}_q]$. If $\mathbf{v}_1, \ldots, \mathbf{v}_q$ are orthonormal ($\mathbf{V}^\top\mathbf{V} = \mathbf{I}_q$),

$$P_{\mathcal{L}_q}(\mathbf{x}) = \mathbf{V}\mathbf{V}^\top\mathbf{x} = s_{\mathbf{v}_1}(\mathbf{x})\mathbf{v}_1 + \cdots + s_{\mathbf{v}_q}(\mathbf{x})\mathbf{v}_q,$$

so $\mathbf{V}^\top\mathbf{x} = (s_{\mathbf{v}_1}(\mathbf{x}), \ldots, s_{\mathbf{v}_q}(\mathbf{x}))^\top$ are the coordinates of the projection when the basis vectors are used as coordinate axes of $\mathcal{L}_q$.

- The scatterplot of variables i and j is the projection of the data $\{\mathbf{x}_1, \ldots, \mathbf{x}_n\}$ onto the 2-dimensional subspace spanned by the standard basis vectors $\mathbf{e}_i, \mathbf{e}_j$.
- If different bases span the same subspace, $\text{span}(\mathbf{v}_1, \mathbf{v}_2) = \text{span}(\mathbf{v}_1', \mathbf{v}_2')$ with $(\mathbf{v}_1, \mathbf{v}_2) \neq (\mathbf{v}_1', \mathbf{v}_2')$, the projection of $\mathbf{x}$ is the same but the scores differ: $(s_{\mathbf{v}_1}(\mathbf{x}), s_{\mathbf{v}_2}(\mathbf{x})) \neq (s_{\mathbf{v}_1'}(\mathbf{x}), s_{\mathbf{v}_2'}(\mathbf{x}))$ in general.

Derivation: why $\mathbf{P}_{\mathbf{V}}$ is the orthogonal projection [Supplement]

$\mathbf{P}_{\mathbf{V}}^\top = \mathbf{P}_{\mathbf{V}}$ and $\mathbf{P}_{\mathbf{V}}^2 = \mathbf{V}(\mathbf{V}^\top\mathbf{V})^{-1}\mathbf{V}^\top\mathbf{V}(\mathbf{V}^\top\mathbf{V})^{-1}\mathbf{V}^\top = \mathbf{P}_{\mathbf{V}}$. Also $\mathbf{P}_{\mathbf{V}}\mathbf{x} \in \mathcal{L}_q$, and the residual satisfies $\mathbf{V}^\top(\mathbf{x} - \mathbf{P}_{\mathbf{V}}\mathbf{x}) = \mathbf{0}$, so it is orthogonal to $\mathcal{L}_q$. By Pythagoras, for any $\mathbf{V}\mathbf{c}$, $\|\mathbf{x} - \mathbf{V}\mathbf{c}\|^2 = \|\mathbf{x} - \mathbf{P}_{\mathbf{V}}\mathbf{x}\|^2 + \|\mathbf{P}_{\mathbf{V}}\mathbf{x} - \mathbf{V}\mathbf{c}\|^2 \geq \|\mathbf{x} - \mathbf{P}_{\mathbf{V}}\mathbf{x}\|^2$: $\mathbf{P}_{\mathbf{V}}\mathbf{x}$ is the closest point of $\mathcal{L}_q$ to $\mathbf{x}$.

3.3 Projection onto an affine subspace and decomposition of observations

The affine subspace $\mathcal{L}_q(\bar{\mathbf{x}})$ is $\mathcal{L}_q$ translated by $\bar{\mathbf{x}}$, so the projection of $\mathbf{x}$ onto it is done in two steps (for simplicity let $\mathbf{v}_1, \ldots, \mathbf{v}_q$ be orthonormal):

(1) Translate, then project: translate $\mathbf{x}$ and $\mathcal{L}_q(\bar{\mathbf{x}})$ together by $-\bar{\mathbf{x}}$, $\mathbf{x} \mapsto \mathbf{x} - \bar{\mathbf{x}}$, $\mathcal{L}_q(\bar{\mathbf{x}}) \mapsto \mathcal{L}_q$, and project $\mathbf{x} - \bar{\mathbf{x}}$ onto $\mathcal{L}_q$: $P_{\mathcal{L}_q}(\mathbf{x} - \bar{\mathbf{x}}) = \mathbf{P}_{\mathbf{V}}(\mathbf{x} - \bar{\mathbf{x}}) = s_{\mathbf{v}_1}(\mathbf{x} - \bar{\mathbf{x}})\mathbf{v}_1 + \cdots + s_{\mathbf{v}_q}(\mathbf{x} - \bar{\mathbf{x}})\mathbf{v}_q$.

(2) Translate back by $\bar{\mathbf{x}}$:

$$P_{\mathcal{L}_q(\bar{\mathbf{x}})}(\mathbf{x}) = \bar{\mathbf{x}} + \mathbf{P}_{\mathbf{V}}(\mathbf{x} - \bar{\mathbf{x}}) = \bar{\mathbf{x}} + s_{\mathbf{v}_1}(\mathbf{x} - \bar{\mathbf{x}})\mathbf{v}_1 + \cdots + s_{\mathbf{v}_q}(\mathbf{x} - \bar{\mathbf{x}})\mathbf{v}_q.$$

The Euclidean distance between $\mathbf{x}$ and $\mathbf{y}$ is the length of $\mathbf{x} - \mathbf{y}$:

$$d(\mathbf{x}, \mathbf{y}) = \Big\{\sum_{i=1}^{p}(x_i - y_i)^2\Big\}^{1/2} = \sqrt{(\mathbf{x} - \mathbf{y})^\top(\mathbf{x} - \mathbf{y})} = \|\mathbf{x} - \mathbf{y}\|.$$

An observation splits into its projection onto an affine subspace and the remainder, and by Pythagoras

$$\mathbf{x} = \underbrace{\bar{\mathbf{x}} + \sum_{j=1}^{q} s_{\mathbf{v}_j}(\mathbf{x} - \bar{\mathbf{x}})\mathbf{v}_j}_{P_{\mathcal{L}_q(\bar{\mathbf{x}})}(\mathbf{x})} + \underbrace{r_{\mathcal{L}_q}(\mathbf{x})}_{\text{remainder}}, \qquad d(\mathbf{x}, \bar{\mathbf{x}})^2 = \sum_{j=1}^{q} s_{\mathbf{v}_j}(\mathbf{x} - \bar{\mathbf{x}})^2 + \|r_{\mathcal{L}_q}(\mathbf{x})\|^2.$$

This decomposition is used repeatedly when we discuss dimension reduction (PCA).

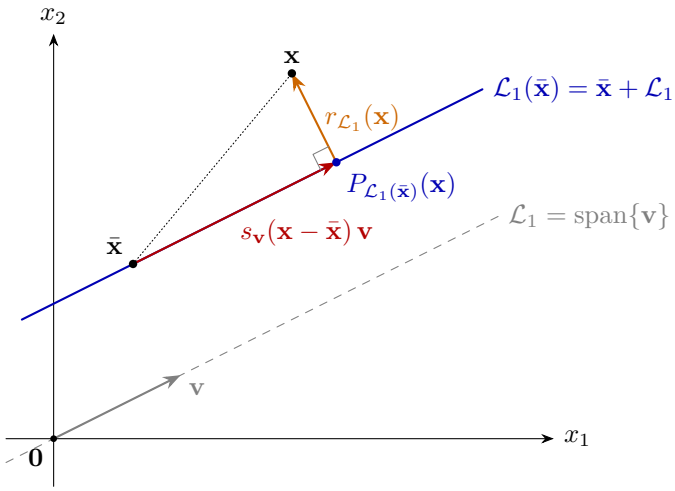


Figure. *Projection onto the affine line $\mathcal{L}_1(\bar{\mathbf{x}})$ with unit direction $\mathbf{v}$: translate by $-\bar{\mathbf{x}}$, project onto the parallel subspace $\mathcal{L}_1$, translate back. The observation splits as $\mathbf{x} = \bar{\mathbf{x}} + s_{\mathbf{v}}(\mathbf{x} - \bar{\mathbf{x}})\mathbf{v} + r_{\mathcal{L}_1}(\mathbf{x})$ with a right angle at the projection, so $d(\mathbf{x}, \bar{\mathbf{x}})^2 = s_{\mathbf{v}}(\mathbf{x} - \bar{\mathbf{x}})^2 + \|r_{\mathcal{L}_1}(\mathbf{x})\|^2$ (dotted side).*

Remark 1.10: geometric meaning of variance

Projection scores are related to conditional variances. If $X_1 \sim N(0, \sigma_1^2)$ and $X_2 = X_1 + \epsilon_2$ with independent $\epsilon_2 \sim N(0, \sigma_2^2)$, then

$$\mathrm{Var}(X_2 \mid X_1) = \mathrm{Var}(\epsilon_2) = \sigma_2^2, \qquad \mathrm{Var}(X_1 \mid X_2) = \frac{\sigma_1^2 \sigma_2^2}{\sigma_1^2 + \sigma_2^2}.$$

[Supplement] The second formula is the conditional-variance formula of Ch. 3, $\mathrm{Var}(X_1 \mid X_2) = \mathrm{Var}(X_1) - \mathrm{Cov}(X_1, X_2)^2 / \mathrm{Var}(X_2) = \sigma_1^2 - \sigma_1^4/(\sigma_1^2 + \sigma_2^2)$: what remains after removing the part of X_1 explained by (projected on) X_2 is the conditional variance.

3.4 Linear and affine transformations

A $q \times p$ matrix $\mathbf{L}$ defines the **linear transformation** $\mathbf{x} \mapsto \mathbf{L}\mathbf{x} \in \mathbb{R}^q$ of $\mathbf{x} \in \mathbb{R}^p$. In general $q > p$ only increases the dimension without adding information, so it is rarely used in multivariate statistics. Examples:

- the orthogonal projection $\mathbf{P}_{\mathbf{V}}\mathbf{x}$ is the linear transformation $\mathbf{x} \mapsto \mathbf{P}_{\mathbf{V}}\mathbf{x} \in \mathbb{R}^p$ by the $p \times p$ matrix $\mathbf{P}_{\mathbf{V}}$;
- the projection scores $(s_{\mathbf{v}_1}(\mathbf{x}), \ldots, s_{\mathbf{v}_q}(\mathbf{x}))$ are the linear transformation $\mathbf{x} \mapsto \mathbf{S}_{\mathbf{V}}\mathbf{x} \in \mathbb{R}^q$ by the $q \times p$ matrix $\mathbf{S}_{\mathbf{V}} = (\mathbf{v}_1/\|\mathbf{v}_1\|, \ldots, \mathbf{v}_q/\|\mathbf{v}_q\|)^\top$. In the simplest case, $\mathbf{x} = \frac{\mathbf{v}_1\mathbf{v}_1^\top}{\|\mathbf{v}_1\|^2}\mathbf{x} + r(\mathbf{x})$ and $s_{\mathbf{v}_1}(\mathbf{x}) = \mathbf{v}_1^\top \mathbf{x}/\|\mathbf{v}_1\|$.

The projection (or the projection scores) onto an affine subspace is not a linear transformation, because of the translation. It belongs to the similar class of **affine transformations**: for a $q \times p$ matrix $\mathbf{L}$ and $\boldsymbol{m} \in \mathbb{R}^q$, $\mathbf{x} \mapsto \boldsymbol{m} + \mathbf{L}\mathbf{x}$.

Theorem 1.11: affine maps and standardization (slides Exercise 1.2)

1. Orthogonal projection onto $\mathcal{L}_q(\bar{\mathbf{x}})$ is an affine transformation, with $\boldsymbol{m} = \bar{\mathbf{x}} - \mathbf{P}_{\mathbf{V}}\bar{\mathbf{x}}$ and $\mathbf{L} = \mathbf{P}_{\mathbf{V}} = \mathbf{V}(\mathbf{V}^\top\mathbf{V})^{-1}\mathbf{V}^\top$; so is the score map $\mathbf{x} \mapsto (s_{\mathbf{v}_1}(\mathbf{x} - \bar{\mathbf{x}}), \ldots, s_{\mathbf{v}_q}(\mathbf{x} - \bar{\mathbf{x}}))^\top$.
2. (Standardization) Let the data have mean vector $\bar{\mathbf{x}}$, covariance matrix $\mathbf{S}$ and $\mathbf{D} =$

$\mathrm{diag}(s_{11},\dots,s_{pp})$. The standardized observations $\mathbf{x}_i' = \mathbf{D}^{-1/2}(\mathbf{x}_i - \bar{\mathbf{x}})$ have sample mean $\mathbf{0}$ and all sample variances equal to 1, and $\mathbf{x} \mapsto \mathbf{D}^{-1/2}(\mathbf{x} - \bar{\mathbf{x}})$ is an affine transformation.

Proof. (1) $P_{\mathcal{L}_q(\bar{\mathbf{x}})}(\mathbf{x}) = \bar{\mathbf{x}} + \mathbf{V}(\mathbf{V}^\top\mathbf{V})^{-1}\mathbf{V}^\top(\mathbf{x} - \bar{\mathbf{x}})$, so $\boldsymbol{m} = \bar{\mathbf{x}} - \mathbf{V}(\mathbf{V}^\top\mathbf{V})^{-1}\mathbf{V}^\top\bar{\mathbf{x}}$ and $\mathbf{L} = \mathbf{V}(\mathbf{V}^\top\mathbf{V})^{-1}\mathbf{V}^\top$. The score vector is $\mathbf{S_V x} - \mathbf{S_V}\bar{\mathbf{x}}$, also affine.

(2) The mean of the transformed data is $\bar{\mathbf{x}}' = \frac{1}{n}\sum_i \mathbf{D}^{-1/2}(\mathbf{x}_i - \bar{\mathbf{x}}) = \mathbf{D}^{-1/2}(\frac{1}{n}\sum_i(\mathbf{x}_i - \bar{\mathbf{x}})) = \mathbf{D}^{-1/2}\cdot\mathbf{0} = \mathbf{0}$. The sample variance of variable j is $S'_{jj} = \mathrm{Var}(s_{jj}^{-1/2}(x_j - \bar{x}_j)) = s_{jj}^{-1/2}s_{jj}s_{jj}^{-1/2} = 1$. Finally $\mathbf{x}' = \mathbf{D}^{-1/2}\mathbf{x} - \mathbf{D}^{-1/2}\bar{\mathbf{x}} = \mathbf{L}\mathbf{x} + \boldsymbol{m}$. ■

3.5 Rotation of coordinates

A linear transformation often used in multivariate statistics is the **linear transformation induced by a change of coordinates**. The usual coordinates of $\mathbb{R}^p$ are given by the standard basis $(\mathbf{e}_1,\dots,\mathbf{e}_p)$ of mutually orthogonal unit vectors. For $p = 2$, rotating the standard coordinate axes gives new coordinates: for an angle θ (radians) let

$$\mathbf{R}_\theta = \begin{pmatrix} \cos\theta & -\sin\theta \\ \sin\theta & \cos\theta \end{pmatrix}, \qquad \mathbf{v}_1 = \mathbf{R}_\theta\mathbf{e}_1, \quad \mathbf{v}_2 = \mathbf{R}_\theta\mathbf{e}_2,$$

the basis rotated counterclockwise by θ. Then $\mathbf{V} = [\mathbf{v}_1, \mathbf{v}_2]$ satisfies $\mathbf{V}^\top\mathbf{V} = \mathbf{V}\mathbf{V}^\top = \mathbf{R}_\theta\mathbf{R}_\theta^\top = \mathbf{I}_2$. An observation $\mathbf{x} = (x_1, x_2)^\top$ is

$$\mathbf{x} = x_1\mathbf{e}_1 + x_2\mathbf{e}_2 = [\mathbf{e}_1, \mathbf{e}_2]\begin{pmatrix} x_1 \\ x_2 \end{pmatrix}, \qquad \mathbf{x} = \mathbf{V}\mathbf{V}^\top\mathbf{x} = [\mathbf{v}_1, \mathbf{v}_2]\begin{pmatrix} \mathbf{v}_1^\top\mathbf{x} \\ \mathbf{v}_2^\top\mathbf{x} \end{pmatrix}.$$

The coordinates of $\mathbf{x}$ are (x_1, x_2) in the standard system and $(\mathbf{v}_1^\top\mathbf{x}, \mathbf{v}_2^\top\mathbf{x})$ in the new system $(\mathbf{v}_1, \mathbf{v}_2)$; the latter are the projection scores onto $\mathrm{span}(\mathbf{v}_1)$ and $\mathrm{span}(\mathbf{v}_2)$. In $\mathbb{R}^p$ one may use any orthonormal basis $(\mathbf{v}_1,\dots,\mathbf{v}_p)$ instead of $(\mathbf{e}_1,\dots,\mathbf{e}_p)$: a linear map by an orthonormal basis is a rotation (change) of coordinates.

4 Standardization, Sphering and Mahalanobis Distance

4.1 Univariate standardization and its limitation

For univariate data (see the data-ellipse discussion above): how far is a point $x \in \mathbb{R}$ from the center μ of the normal distribution $N(\mu, \sigma^2)$? The **standardized distance** $|x - \mu|/\sigma$ is more meaningful than the difference $x - \mu$. For univariate data $x_1,\dots,x_n$ with mean $\bar{x}$ and standard deviation s, the **standardization**

$$x_i \mapsto \frac{x_i - \bar{x}}{s}$$

gives data with mean 0 and standard deviation 1.

Standardization of multivariate data makes the mean and variance of every variable 0 and 1: with $\mathbf{D} = \mathrm{diag}(s_1^2,\dots,s_p^2)$, the naive transformation is $\mathbf{x} \mapsto \mathbf{D}^{-1/2}(\mathbf{x} - \bar{\mathbf{x}})$. It is used often, to keep results from depending on the units of measurement. But unlike the univariate case, standardization alone **does not give a distance that reflects the distribution**, because it ignores the correlations.

Example 1.12: distances of points in bivariate data (slides Example 1.3)

For bivariate data with $\bar{\mathbf{x}} = \mathbf{0}$, which of the points $\mathbf{y} = (-500, 10)^\top$ and $\mathbf{z} = (1500, 15)^\top$ is farther from the center? The Euclidean distances are $d(\bar{\mathbf{x}}, \mathbf{y}) = \sqrt{500^2 + 10^2} = 500.1$ and $d(\bar{\mathbf{x}}, \mathbf{z}) = \sqrt{1500^2 + 15^2} = 1500.075$, so $\mathbf{z}$ is farther. Yet in the scatterplot there are many observations around $\mathbf{z}$ and almost none around $\mathbf{y}$: relative to the distribution, $\mathbf{y}$ looks farther from the center.

The sample covariance is $\mathbf{S} = \begin{pmatrix} 640000 & 9600 \\ 9600 & 225 \end{pmatrix} = \begin{pmatrix} 800^2 & 800 \cdot 15 \cdot 0.8 \\ 800 \cdot 15 \cdot 0.8 & 15^2 \end{pmatrix}$ (correlation 0.8). After standardization $\mathbf{y}' = \mathbf{D}^{-1/2}(\mathbf{y} - \mathbf{0}) = (-0.625, 0.667)^\top$ and $\mathbf{z}' = \mathbf{D}^{-1/2}(\mathbf{z} - \mathbf{0}) = (1.875, 1.000)^\top$, and still $d(\mathbf{0}, \mathbf{y}') = 0.91 < 2.125 = d(\mathbf{0}, \mathbf{z}')$: the standardized $\mathbf{y}'$ is closer to the center. Each variable now has standard deviation 1, but the standard deviation in the direction of $\mathbf{y}'$ is $sd(s_{\mathbf{y}'}(\mathbf{X}')) = 0.36$ and in the direction of $\mathbf{z}'$ it is $sd(s_{\mathbf{z}'}(\mathbf{X}')) = 1.31$, not 1. Standardization does not standardize the spread in an arbitrary direction.

4.2 Sphering

The spread of the data is described by the covariance matrix $\mathbf{S}$ and the corresponding ellipse (ellipsoid). If the covariance is the identity $\mathbf{I}_p$, the ellipse is the set of $\mathbf{x} \in \mathbb{R}^p$ with $c = (\mathbf{x} - \bar{\mathbf{x}})^\top \mathbf{I}_p^{-1}(\mathbf{x} - \bar{\mathbf{x}}) = \|\mathbf{x} - \bar{\mathbf{x}}\|^2$: the distribution is shown as a sphere, and the spread is 1 in **every** direction $\mathbf{v}$. A **sphering transformation** is a transformation after which the covariance matrix is the identity.

Theorem 1.13: isotropy of sphered data (slides homework)

If the data $\mathbf{X}$ have mean vector $\mathbf{0}$ and covariance matrix $\mathbf{I}_p$, then for any unit vector $\mathbf{v}$ the scores $S_{\mathbf{v}}(\mathbf{X}) = [S_{\mathbf{v}}(\mathbf{x}_1), \ldots, S_{\mathbf{v}}(\mathbf{x}_n)]$, $S_{\mathbf{v}}(\mathbf{x}_i) = \mathbf{v}^\top \mathbf{x}_i$, have sample variance 1.

Proof. The scores have mean $\mathbf{v}^\top \bar{\mathbf{x}} = 0$ and sample variance $\frac{1}{n-1}\sum_i (\mathbf{v}^\top \mathbf{x}_i)^2 = \mathbf{v}^\top \mathbf{S} \mathbf{v} = \mathbf{v}^\top \mathbf{I}_p \mathbf{v} = \|\mathbf{v}\|^2 = 1$. ■

Example 1.14: sphering a diagonal covariance (slides Example 1.5)

Let $\mathbf{X} = [\mathbf{x}_1, \ldots, \mathbf{x}_n]^\top \in \mathbb{R}^{n \times 2}$ have $\bar{\mathbf{x}} = (10, 10)^\top$ and $\mathbf{S} = \begin{pmatrix} 4 & 0 \\ 0 & 9 \end{pmatrix}$. The standardization matrix that makes each variance 1 is $\mathbf{D}^{-1/2} = \mathrm{diag}(1/2, 1/3)$, and $\mathbf{x} \mapsto \mathbf{D}^{-1/2}\mathbf{x}$ is the sphering transformation of these data: the transformed covariance is

$$\mathbf{S}' = \frac{1}{n-1}\sum_{i=1}^{n} (\mathbf{D}^{-1/2}\mathbf{x}_i - \mathbf{D}^{-1/2}\bar{\mathbf{x}})(\mathbf{D}^{-1/2}\mathbf{x}_i - \mathbf{D}^{-1/2}\bar{\mathbf{x}})^\top$$

$$\mathbf{S}' = \frac{1}{n-1}\sum_{i=1}^{n} \mathbf{D}^{-1/2}(\mathbf{x}_i - \bar{\mathbf{x}})(\mathbf{x}_i - \bar{\mathbf{x}})^\top \mathbf{D}^{-1/2} = \mathbf{D}^{-1/2}\mathbf{S}\mathbf{D}^{-1/2} = \mathbf{D}^{-1/2}\mathbf{D}\mathbf{D}^{-1/2} = \mathbf{I}_2.$$

Sphering turns the elliptical covariance structure into a sphere.

Definition 1.15: sphering for a general covariance matrix

For a general $\mathbf{S}$, first rotate the coordinates so that all pairwise sample correlations become 0, then standardize. Given $\mathbf{X}$ with covariance $\mathbf{S}$, the sphering transformation is built in three steps.

1. **Rotation**: choose an orthonormal basis matrix $\mathbf{U}_S = [\mathbf{u}_1, \dots, \mathbf{u}_p]$ such that the rotated observations $\mathbf{x}'_i = \mathbf{U}_S^\top \mathbf{x}_i$ $(i = 1, \dots, n)$ have a diagonal sample covariance $\mathbf{D}_S = (d_{jk})$ $(d_{jk} = 0$ for $j \neq k)$, i.e. $\mathbf{D}_S = \mathbf{U}_S^\top \mathbf{S} \mathbf{U}_S = \operatorname{diag}(d_{11}, \dots, d_{pp})$.
2. **Standardization**: with $\mathbf{D}_S^{-1/2} = \operatorname{diag}(d_{11}^{-1/2}, \dots, d_{pp}^{-1/2})$, apply $\mathbf{x}' \mapsto \mathbf{D}_S^{-1/2} \mathbf{x}'$ to the rotated observations.
3. **Back-rotation**: rotate back by the inverse of the first rotation, $\mathbf{x} \mapsto \mathbf{U}_S \mathbf{x}$.

Applying the three steps in turn gives the sphering transformation for $\mathbf{S}$:

$$\mathbf{x} \mapsto \mathbf{U}_S \mathbf{D}_S^{-1/2} \mathbf{U}_S^\top \mathbf{x}.$$

[Supplement] The sphered covariance is $\mathbf{I}_p$. By the first step $\mathbf{S} = \mathbf{U}_S \mathbf{D}_S \mathbf{U}_S^\top$, so

$$\begin{aligned}
\mathbf{S}' &= \tfrac{1}{n-1} \textstyle\sum_i \mathbf{U}_S \mathbf{D}_S^{-1/2} \mathbf{U}_S^\top (\mathbf{x}_i - \bar{\mathbf{x}})(\mathbf{x}_i - \bar{\mathbf{x}})^\top \mathbf{U}_S \mathbf{D}_S^{-1/2} \mathbf{U}_S^\top \\
&= \mathbf{U}_S \mathbf{D}_S^{-1/2} \mathbf{U}_S^\top \, \mathbf{S} \, \mathbf{U}_S \mathbf{D}_S^{-1/2} \mathbf{U}_S^\top \\
&= \mathbf{U}_S \mathbf{D}_S^{-1/2} \mathbf{U}_S^\top \, \mathbf{U}_S \mathbf{D}_S \mathbf{U}_S^\top \, \mathbf{U}_S \mathbf{D}_S^{-1/2} \mathbf{U}_S^\top = \mathbf{U}_S \mathbf{D}_S^{-1/2} \mathbf{D}_S \mathbf{D}_S^{-1/2} \mathbf{U}_S^\top = \mathbf{U}_S \mathbf{U}_S^\top = \mathbf{I}_p.
\end{aligned}$$

■

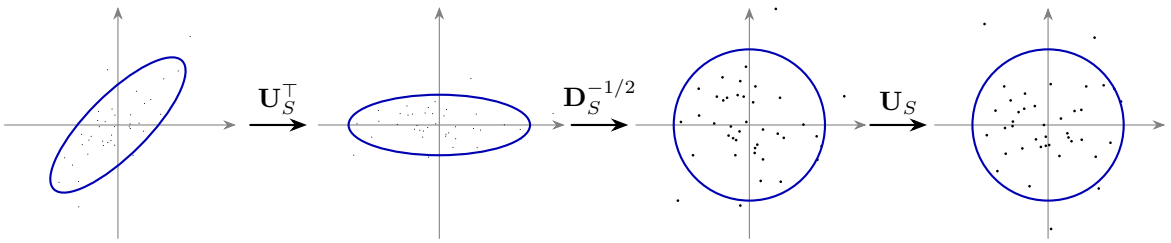
Example 1.16: applying the sphering transformation (slides Example 1.6)

Find the sphering transformation for data $\mathbf{X} \in \mathbb{R}^{n \times 2}$ with $\bar{\mathbf{x}} = \mathbf{0}$ and $\mathbf{S} = \begin{pmatrix} 5 & 4 \\ 4 & 5 \end{pmatrix}$.

First rotate with the orthonormal basis matrix $\mathbf{U}_S = \frac{1}{\sqrt{2}} \begin{pmatrix} 1 & -1 \\ 1 & 1 \end{pmatrix}$, for which the rotated observations have diagonal sample covariance $\mathbf{D}_S = \operatorname{diag}(9, 1)$:

$$\mathbf{U}_S^\top \mathbf{S} \mathbf{U}_S = \frac{1}{\sqrt{2}} \begin{pmatrix} 1 & 1 \\ -1 & 1 \end{pmatrix} \begin{pmatrix} 5 & 4 \\ 4 & 5 \end{pmatrix} \frac{1}{\sqrt{2}} \begin{pmatrix} 1 & -1 \\ 1 & 1 \end{pmatrix} = \begin{pmatrix} 9 & 0 \\ 0 & 1 \end{pmatrix} = \mathbf{D}_S;$$

then standardize and rotate back. Note that $\mathbf{U}_S = \begin{pmatrix} \cos\theta & -\sin\theta \\ \sin\theta & \cos\theta \end{pmatrix}$ with $\theta = \pi/4$. Hence the sphering transformation is $\mathbf{x} \mapsto \mathbf{U}_S \operatorname{diag}(1/3, 1) \mathbf{U}_S^\top \mathbf{x}$.



(a) data, cov $\mathbf{S}$ (b) rotated, cov $\operatorname{diag}(9, 1)$ (c) standardized, cov $\mathbf{I}_2$ (d) rotated back, cov $\mathbf{I}_2$

Figure. *The three steps of sphering for* $\mathbf{S}$ *of Example 1.16, applied to 40 points with exactly this sample covariance (computed in Python): (a) original data with ellipse* $\mathbf{x}^\top \mathbf{S}^{-1} \mathbf{x} = 4$*; (b) rotation by* $45°$ *clockwise,* $\mathbf{x} \mapsto \mathbf{U}_S^\top \mathbf{x}$*; (c) standardization; (d) rotation back by* $45°$ *counterclockwise. The overall map is* $\mathbf{S}^{-1/2} = \mathbf{U}_S \mathbf{D}_S^{-1/2} \mathbf{U}_S^\top$ *and the ellipse becomes a circle.*

Applied with the sample covariance matrix of the data of Example 1.12, the sphering transformation shows the "outlier-looking" point $\mathbf{y}$ as farther from the center than $\mathbf{z}$: the Euclidean distances of

the sphered points $\mathbf{y}'$, $\mathbf{z}'$ to the sphered center $\bar{\mathbf{x}}'$ are

$$\|\mathbf{y}' - \bar{\mathbf{x}}'\| = 2.65 > 2.18 = \|\mathbf{z}' - \bar{\mathbf{x}}'\|.$$

[Supplement] These two numbers come from the covariance matrix of the plotted sample. With the rounded matrix $\mathbf{S}$ quoted in Example 1.12, the same computation gives 2.04 and 2.05 (a Python check): $\mathbf{y}$, a third as far as $\mathbf{z}$ in Euclidean distance, is then as far as $\mathbf{z}$ in Mahalanobis distance. Either way, the distribution-based distance reverses the misleading Euclidean comparison.

4.3 Mahalanobis distance

A distance that takes the full information in the covariance matrix into account is the **Mahalanobis distance**, an example of a statistical distance that reflects the distribution.

Definition 1.17: Mahalanobis (statistical) distance

For a distribution with mean vector $\bar{\mathbf{x}}$ and covariance matrix $\mathbf{S}$, the Mahalanobis distance of a point $\mathbf{y}$ from the mean $\bar{\mathbf{x}}$ is the Euclidean distance between the sphered points $\mathbf{y}' = \mathbf{U}_S\mathbf{D}_S^{-1/2}\mathbf{U}_S^\top\mathbf{y}$ and $\bar{\mathbf{x}}' = \mathbf{U}_S\mathbf{D}_S^{-1/2}\mathbf{U}_S^\top\bar{\mathbf{x}}$:

$$d_S(\mathbf{y}, \bar{\mathbf{x}}) = \|\mathbf{y}' - \bar{\mathbf{x}}'\| = \|\mathbf{U}_S\mathbf{D}_S^{-1/2}\mathbf{U}_S^\top(\mathbf{y} - \bar{\mathbf{x}})\|.$$

Theorem 1.18: quadratic form of the Mahalanobis distance

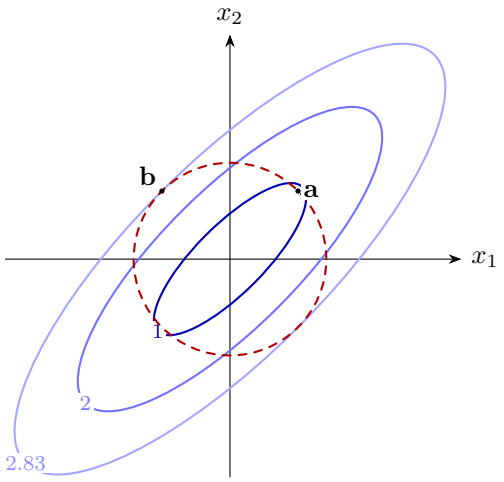
$d_S(\mathbf{y}, \bar{\mathbf{x}})^2 = (\mathbf{y} - \bar{\mathbf{x}})^\top\mathbf{S}^{-1}(\mathbf{y} - \bar{\mathbf{x}})$, and the sphering transformation is $\mathbf{x} \mapsto \mathbf{S}^{-1/2}\mathbf{x}$.

Proof. Using the orthogonality of $\mathbf{U}_S$,

$$d_S(\mathbf{y}, \bar{\mathbf{x}})^2 = (\mathbf{y} - \bar{\mathbf{x}})^\top\mathbf{U}_S\mathbf{D}_S^{-1/2}\underbrace{\mathbf{U}_S^\top\mathbf{U}_S}_{\mathbf{I}_p}\mathbf{D}_S^{-1/2}\mathbf{U}_S^\top(\mathbf{y} - \bar{\mathbf{x}}) = (\mathbf{y} - \bar{\mathbf{x}})^\top\mathbf{U}_S\mathbf{D}_S^{-1}\mathbf{U}_S^\top(\mathbf{y} - \bar{\mathbf{x}}).$$

More simply, $\mathbf{U}_S\mathbf{D}_S^{-1}\mathbf{U}_S^\top = \mathbf{S}^{-1}$, which gives $d_S(\mathbf{y}, \bar{\mathbf{x}})^2 = (\mathbf{y} - \bar{\mathbf{x}})^\top\mathbf{S}^{-1}(\mathbf{y} - \bar{\mathbf{x}})$. Also $\mathbf{U}_S\mathbf{D}_S^{-1/2}\mathbf{U}_S^\top = \mathbf{S}^{-1/2}$, so the sphering transformation can be written $\mathbf{x} \mapsto \mathbf{S}^{-1/2}\mathbf{x}$. ■

The multivariate counterparts of univariate standardization (studentisation) and of the standardized distance are the sphering transformation and the Mahalanobis distance.



$\mathbf{a} = (2, 2)$: $d_S(\mathbf{a}, \mathbf{0}) = 0.94$
$\mathbf{b} = (-2, 2)$: $d_S(\mathbf{b}, \mathbf{0}) = 2.83$
dashed: $\|\mathbf{x}\| = 2.83$
solid: $d_S(\mathbf{x}, \mathbf{0}) = 1, 2, 2.83$

Figure. *Euclidean versus Mahalanobis distance for* $\bar{\mathbf{x}} = \mathbf{0}$ *and* $\mathbf{S} = \begin{pmatrix} 5 & 4 \\ 4 & 5 \end{pmatrix}$. *The points* $\mathbf{a}$ *and* $\mathbf{b}$ *lie on the same Euclidean circle (dashed), but* $\mathbf{a}$ *lies along the long axis of the data ellipses* $d_S = 1, 2, 2.83$ *and* $\mathbf{b}$ *against the correlation, so* $d_S(\mathbf{a}, \mathbf{0}) = 0.94$ *while* $d_S(\mathbf{b}, \mathbf{0}) = 2.83$.

5 Eigendecomposition and Positive Definiteness

Motivation. A covariance matrix is symmetric and positive semidefinite, so it has an orthonormal eigenbasis. The eigenvectors are the natural axes of the data cloud and the eigenvalues the variances along them; PCA, Mahalanobis distance and the normal density all rest on this decomposition.

5.1 Eigendecomposition

For a symmetric $\mathbf{A} \in \mathbb{R}^{p\times p}$, a nonzero $\mathbf{u} \in \mathbb{R}^p$ satisfying $\mathbf{Au} = \lambda\mathbf{u}$ is an **eigenvector** and the scalar $\lambda \in \mathbb{R}$ an **eigenvalue**. The left side is the linear transformation of $\mathbf{u}$ by $\mathbf{A}$, and the right side only stretches or shrinks $\mathbf{u}$ by λ: the eigenvectors of $\mathbf{A}$ keep their direction under $\mathbf{A}$, and the eigenvalues give the factor of stretching.

Example 1.19: eigenstructure of the sphering matrix (slides Example 1.7)

Check the eigendecomposition on the sphering matrix $\mathbf{A} = \begin{pmatrix} 2/3 & -1/3 \\ -1/3 & 2/3 \end{pmatrix}$ of Example 1.16: applying it to the vectors $\mathbf{u}_1 = (1,1)^\top/\sqrt{2}$ and $\mathbf{u}_2 = (-1,1)^\top/\sqrt{2}$ used in the first (rotation) step only rescales their lengths. Indeed

$$\mathbf{Au}_1 = \frac{1}{\sqrt{2}}\begin{pmatrix} \frac{2}{3}\cdot 1 + (-\frac{1}{3})\cdot 1 \\ (-\frac{1}{3})\cdot 1 + \frac{2}{3}\cdot 1 \end{pmatrix} = \frac{1}{\sqrt{2}}\begin{pmatrix} \frac{1}{3} \\ \frac{1}{3} \end{pmatrix} = \frac{1}{3}\mathbf{u}_1.$$

[Supplement] Similarly $\mathbf{Au}_2 = \frac{1}{\sqrt{2}}(-\frac{2}{3}-\frac{1}{3},\ \frac{1}{3}+\frac{2}{3})^\top = \mathbf{u}_2$. Sphering rescales the eigen-direction of eigenvalue λ_j of $\mathbf{S}$ by $1/\sqrt{\lambda_j}$ (1/3 and 1 here).

Theorem 1.20: properties of eigenvalues and eigenvectors (slides homework)

Let the symmetric $\mathbf{A} \in \mathbb{R}^{p\times p}$ have p eigenvalues; for convenience assume they are distinct and ordered, $\lambda_1 > \lambda_2 > \cdots > \lambda_p$.

1. If $\mathbf{u}_i$ is an eigenvector for λ_i, so is $c\mathbf{u}_i$ for every $c \neq 0$: eigenvectors are not unique.
2. Among the vectors of length 1 ($\|\mathbf{u}\| = 1$) there are exactly two eigenvectors for λ_i, $\mathbf{u}_i$ and $-\mathbf{u}_i$. Ignoring the sign, the unit eigenvector $\mathbf{u}_i$ for λ_i is unique.
3. Eigenvectors $\mathbf{u}_i, \mathbf{u}_j$ for distinct eigenvalues $\lambda_i \neq \lambda_j$ are orthogonal: $\mathbf{u}_i^\top\mathbf{u}_j = 0$.

Proof of (3). Given $\mathbf{Au}_i = \lambda_i\mathbf{u}_i$ and $\mathbf{Au}_j = \lambda_j\mathbf{u}_j$, compute $\mathbf{u}_i^\top\mathbf{Au}_j$ in two ways. First, $\mathbf{u}_i^\top\mathbf{Au}_j = \mathbf{u}_i^\top(\lambda_j\mathbf{u}_j) = \lambda_j(\mathbf{u}_i^\top\mathbf{u}_j)$. Second, $\mathbf{A}^\top = \mathbf{A}$, so transposing $\mathbf{Au}_i = \lambda_i\mathbf{u}_i$ gives $\mathbf{u}_i^\top\mathbf{A} = \lambda_i\mathbf{u}_i^\top$; multiplying on the right by $\mathbf{u}_j$, $\mathbf{u}_i^\top\mathbf{Au}_j = \lambda_i(\mathbf{u}_i^\top\mathbf{u}_j)$. Hence $\lambda_i(\mathbf{u}_i^\top\mathbf{u}_j) = \lambda_j(\mathbf{u}_i^\top\mathbf{u}_j)$, and since $\lambda_i \neq \lambda_j$, $\mathbf{u}_i^\top\mathbf{u}_j = 0$. ■

Theorem 1.21: eigendecomposition of a symmetric matrix (slides homework)

Let the eigenvalue–eigenvector pairs of the symmetric $\mathbf{A} \in \mathbb{R}^{p\times p}$ be $(\lambda_i, \mathbf{u}_i)$, $i = 1, \ldots, p$, with $\|\mathbf{u}_i\| = 1$. The eigendecomposition of $\mathbf{A}$ is

$$\mathbf{A} = \mathbf{U}\mathbf{\Lambda}\mathbf{U}^\top = \sum_{i=1}^{p} \lambda_i\mathbf{u}_i\mathbf{u}_i^\top,$$

where the eigenvector matrix $\mathbf{U} = [\mathbf{u}_1, \ldots, \mathbf{u}_p]$ is orthogonal and the eigenvalue matrix $\mathbf{\Lambda} = \operatorname{diag}(\lambda_1, \ldots, \lambda_p)$ is diagonal.

The most important matrices in multivariate statistics are symmetric covariance matrices $\mathbf{S}$, so we restrict the eigendecomposition to symmetric matrices. It is not unique in general; whether it is depends on the eigenvalues.

- If the p eigenvalues are distinct, $\lambda_1 > \lambda_2 > \cdots > \lambda_p$, the decomposition $\mathbf{U\Lambda U}^\top$ is unique (up to the signs of the eigenvectors: $\mathbf{u}_i$ and $-\mathbf{u}_i$ are the same eigenvector).
- If the eigenvalues are not distinct, an eigendecomposition still exists, but the eigenvector matrix is not unique. For example, $\sigma^2\mathbf{I}_p$ has eigenvalues $\lambda_i = \sigma^2$ for all i, and its eigenvector matrix can be any orthogonal $\mathbf{U}$ ($\mathbf{UU}^\top = \mathbf{I}_p$).
- For a given matrix the eigendecomposition generally has no closed-form solution; it is computed numerically, e.g. by orthogonal iteration.

5.2 Inverse, definiteness, square root

The inverse of a symmetric matrix can be obtained from its eigendecomposition:

$$\mathbf{A}^{-1} = \mathbf{U\Lambda}^{-1}\mathbf{U}^\top, \qquad \mathbf{\Lambda}^{-1} = \operatorname{diag}\Big(\frac{1}{\lambda_1}, \ldots, \frac{1}{\lambda_p}\Big),$$

defined only when no eigenvalue is 0 ($\lambda_i \neq 0$). A symmetric $\mathbf{A}$ whose eigenvalues are all positive ($\lambda_i > 0$) is called **positive definite** (PD); one whose eigenvalues are all non-negative ($\lambda_i \geq 0$) is called **non-negative definite** (NND; positive semi-definite).

Definition 1.22: NND and PD

The definiteness of a matrix $\mathbf{A} \in \mathbb{R}^{p\times p}$ is defined as follows.

1. (NND) If $\mathbf{x}^\top\mathbf{Ax} \geq 0$ for all $\mathbf{x} \in \mathbb{R}^p$, $\mathbf{A}$ is non-negative definite.
2. (PD) If $\mathbf{x}^\top\mathbf{Ax} > 0$ for all nonzero $\mathbf{x} \in \mathbb{R}^p$, $\mathbf{A}$ is positive definite.

Theorem 1.23: eigenvalues and definiteness (slides homework)

1. If all eigenvalues of a symmetric $\mathbf{A}$ are positive, $\mathbf{A}$ is PD.
2. If all eigenvalues of a symmetric $\mathbf{A}$ are non-negative, $\mathbf{A}$ is NND.

Proof. We must show $\mathbf{x}^\top\mathbf{Ax} > 0$ for every nonzero $\mathbf{x} \in \mathbb{R}^p$.

Step 1 (eigendecomposition). $\mathbf{A}$ is symmetric, so $\mathbf{A} = \mathbf{U\Lambda U}^\top$ with $\mathbf{U}$ orthogonal and $\mathbf{\Lambda} = \operatorname{diag}(\lambda_1, \ldots, \lambda_p)$.

Step 2 (quadratic form). $\mathbf{x}^\top\mathbf{Ax} = \mathbf{x}^\top\mathbf{U\Lambda U}^\top\mathbf{x}$.

Step 3 (substitution). Let $\mathbf{y} = \mathbf{U}^\top\mathbf{x}$. Since $\mathbf{U}^\top\mathbf{U} = \mathbf{I}$, $\mathbf{y}^\top\mathbf{y} = \mathbf{x}^\top\mathbf{x}$, and $\mathbf{x}^\top\mathbf{Ax} = \mathbf{y}^\top\mathbf{\Lambda y} = \sum_{i=1}^p \lambda_i y_i^2$.

Step 4 (positivity). Since $\lambda_i > 0$ and $y_i^2 \geq 0$, and $\mathbf{x} \neq \mathbf{0}$ implies $\mathbf{y} \neq \mathbf{0}$, at least one $y_i \neq 0$; hence $\sum_i \lambda_i y_i^2 > 0$ and $\mathbf{A}$ is PD.

For (2) the same computation gives $\sum_i \lambda_i y_i^2 \geq 0$. ■

[Supplement] The converses also hold: if $\mathbf{A}$ is PD (NND), then $\lambda_i = \mathbf{u}_i^\top \mathbf{A}\mathbf{u}_i > 0$ (≥ 0), using $\mathbf{A}\mathbf{u}_i = \lambda_i \mathbf{u}_i$ and $\|\mathbf{u}_i\| = 1$. So a symmetric matrix is PD (NND) if and only if all its eigenvalues are positive (non-negative).

- If a symmetric $\mathbf{A}$ is PD, its **inverse exists**. When the sample size n is much larger than the dimension p, the sample covariance matrix $\mathbf{S}$ is PD in "most" cases, so $\mathbf{S}^{-1}$ exists.
- In general the sample covariance matrix is not always PD, but it is always NND.
- An NND matrix that is not PD has a zero eigenvalue, so **its inverse does not exist**.

Theorem 1.24: the sample covariance is always NND, not necessarily PD (slides homework)

Proof. Step 1 (NND). Substituting $\mathbf{S}$, $\mathbf{z}^\top \mathbf{S}\mathbf{z} = \mathbf{z}^\top \Big(\frac{1}{n-1} \sum_{i=1}^n (\mathbf{x}_i - \bar{\mathbf{x}})(\mathbf{x}_i - \bar{\mathbf{x}})^\top \Big) \mathbf{z} = \frac{1}{n-1} \sum_{i=1}^n \{(\mathbf{x}_i - \bar{\mathbf{x}})^\top \mathbf{z}\}^2$. Each $(\mathbf{x}_i - \bar{\mathbf{x}})^\top \mathbf{z}$ is a scalar, so its square is non-negative and so is the sum: $\mathbf{z}^\top \mathbf{S}\mathbf{z} \geq 0$ for every $\mathbf{z} \in \mathbb{R}^p$.

Step 2 (not necessarily PD). For $\mathbf{S}$ to be PD we would need $\mathbf{z}^\top \mathbf{S}\mathbf{z} > 0$ for all nonzero $\mathbf{z}$. If the data points $\mathbf{x}_1, \ldots, \mathbf{x}_n$ lie in a lower-dimensional subspace of $\mathbb{R}^p$ (e.g. all points on a line or in a plane within $\mathbb{R}^p$), $\mathbf{S}$ has at least one zero eigenvalue, and there is a nonzero $\mathbf{z}$ with $\mathbf{z}^\top \mathbf{S}\mathbf{z} = 0$. So $\mathbf{S}$ is always NND, but not necessarily PD: it can have zero eigenvalues when the points do not spread out in all p dimensions. ■

Definition 1.25: symmetric square root

For NND $\mathbf{S} = \mathbf{U}\mathbf{\Lambda}\mathbf{U}^\top$, define $\mathbf{S}^{1/2} = \mathbf{U}\mathbf{\Lambda}^{1/2}\mathbf{U}^\top$, $\mathbf{\Lambda}^{1/2} = \operatorname{diag}(\sqrt{\lambda_1}, \ldots, \sqrt{\lambda_p})$. The square root $\mathbf{S}^{1/2}$ so defined is symmetric and satisfies $\mathbf{S}^{1/2}\mathbf{S}^{1/2} = \mathbf{S}$. Likewise for symmetric PD $\mathbf{S}$ the negative square root is

$$\mathbf{S}^{-1/2} = (\mathbf{S}^{-1})^{1/2} = \mathbf{U}\mathbf{\Lambda}^{-1/2}\mathbf{U}^\top, \qquad \mathbf{\Lambda}^{-1/2} = \operatorname{diag}\Big(\frac{1}{\sqrt{\lambda_1}}, \ldots, \frac{1}{\sqrt{\lambda_p}}\Big).$$

5.3 Covariance matrices and ellipses

The shape of a covariance matrix $\mathbf{S}$ can be described by its ellipse $E_S = \{\mathbf{x} \in \mathbb{R}^p : \mathbf{x}^\top \mathbf{S}^{-1} \mathbf{x} = 1\}$. The shape of this ellipse is closely tied to the eigendecomposition: with $\mathbf{S} = \mathbf{U}\mathbf{\Lambda}\mathbf{U}^\top$,

$$\frac{(\mathbf{u}_1^\top \mathbf{x})^2}{\lambda_1} + \cdots + \frac{(\mathbf{u}_p^\top \mathbf{x})^2}{\lambda_p} = 1,$$

so the principal axes of E_S are parallel to the eigenvectors $\mathbf{u}_i$, and the half-length of each axis is the square root $\sqrt{\lambda_i}$ of the eigenvalue. The eigendecomposition of covariance matrices is one of the most important concepts in multivariate statistics; through $\mathbf{S} = \mathbf{U}\mathbf{\Lambda}\mathbf{U}^\top$ we have

1. the change to coordinates in which all correlations are 0 ($\mathbf{x} \mapsto \mathbf{U}^\top \mathbf{x}$), and
2. the sphering transformation to covariance $\mathbf{I}_p$ ($\mathbf{x} \mapsto \mathbf{U}\mathbf{\Lambda}^{-1/2}\mathbf{U}^\top \mathbf{x}$).

Theorem 1.26: covariance of rotated data (slides Discussion)

If the covariance matrix of $\mathbf{X}$ is $\mathbf{S} = \mathbf{U}\mathbf{\Lambda}\mathbf{U}^\top$, the rotated data $\mathbf{x}_i' = \mathbf{U}^\top \mathbf{x}_i$, $i = 1, \ldots, n$, have sample covariance matrix equal to the eigenvalue matrix $\mathbf{\Lambda}$. The eigenvalues of a covariance matrix are therefore the variances of mutually uncorrelated rotated coordinates.

[Supplement]. The rotated mean is $\mathbf{U}^\top \bar{\mathbf{x}}$, so the sample covariance is $\frac{1}{n-1}\sum_i \mathbf{U}^\top(\mathbf{x}_i - \bar{\mathbf{x}})(\mathbf{x}_i - \bar{\mathbf{x}})^\top \mathbf{U} = \mathbf{U}^\top \mathbf{S}\mathbf{U} = \mathbf{U}^\top \mathbf{U}\mathbf{\Lambda}\mathbf{U}^\top \mathbf{U} = \mathbf{\Lambda}$. ■

6 Geometry of Principal Component Analysis

To explore data with many variables, displaying and inspecting the scatterplots of all pairs of variables is inefficient. We therefore extract q **principal components** from the data and study the main variation through a few new variables. **Principal component analysis** (PCA) finds, in order, the linear combinations of the original variables that explain the most variation of the data. Here we obtain the principal components from the eigendecomposition of the covariance matrix and study the geometric meaning of projection and dimension reduction and its visualization. PCA is a representative dimension-reduction method and is treated again in Ch. 7.

6.1 Which direction to project on?

Center the observation $\mathbf{x}_i$ at the mean $\bar{\mathbf{x}}$ and project onto a unit vector $\mathbf{a}$; the score is $t_i(\mathbf{a}) = \mathbf{a}^\top(\mathbf{x}_i - \bar{\mathbf{x}})$, $\|\mathbf{a}\| = 1$. The sample variance of these projection scores is

$$\frac{1}{n-1}\sum_{i=1}^{n} t_i(\mathbf{a})^2 = \mathbf{a}^\top \mathbf{S}\mathbf{a}.$$

Definition 1.27: principal component directions

The first principal component direction is the direction that maximizes the variance of the projection scores: $\mathbf{u}_1 \in \arg\max_{\|\mathbf{a}\|=1} \mathbf{a}^\top \mathbf{S}\mathbf{a}$. The optimal direction is the eigenvector for the largest eigenvalue of $\mathbf{S}$. Later principal component directions maximize the variance among directions orthogonal to those already found.

[Supplement] $\mathbf{u}_1$ is the top eigenvector. With $\mathbf{z} = \mathbf{U}^\top \mathbf{a}$ ($\|\mathbf{z}\| = 1$), $\mathbf{a}^\top \mathbf{S}\mathbf{a} = \sum_j \lambda_j z_j^2 \le \lambda_1 \sum_j z_j^2 = \lambda_1$, with equality at $\mathbf{a} = \mathbf{u}_1$. Under the constraint $\mathbf{a} \perp \mathbf{u}_1$, i.e. $z_1 = 0$, the maximum is λ_2 at $\mathbf{u}_2$, and so on. ■

6.2 Principal component scores

The ith row of the centered data matrix $\mathbf{X}_C$ is $(\mathbf{x}_i - \bar{\mathbf{x}})^\top$, and the sample covariance is $\mathbf{S} = (n-1)^{-1}\mathbf{X}_C^\top \mathbf{X}_C$. Let $\mathbf{S} = \mathbf{U}\mathbf{\Lambda}\mathbf{U}^\top$ with $\mathbf{U} = [\mathbf{u}_1, \ldots, \mathbf{u}_p]$ orthogonal and $\mathbf{\Lambda} = \mathrm{diag}(\lambda_1, \ldots, \lambda_p)$, $\lambda_1 \ge \cdots \ge \lambda_p \ge 0$. Using the eigenvectors as new coordinate axes gives the principal component score matrix

$$\mathbf{T} = \mathbf{X}_C \mathbf{U} = (t_{ij}), \qquad t_{ij} = (\mathbf{x}_i - \bar{\mathbf{x}})^\top \mathbf{u}_j.$$

Each column of $\mathbf{T}$ has mean 0, and its sample covariance is $\frac{1}{n-1}\mathbf{T}^\top \mathbf{T} = \mathbf{U}^\top \mathbf{S}\mathbf{U} = \mathbf{\Lambda}$. Hence the jth principal component has sample variance λ_j, and different principal component scores are sample-uncorrelated.

Example 1.28: principal components of two test scores

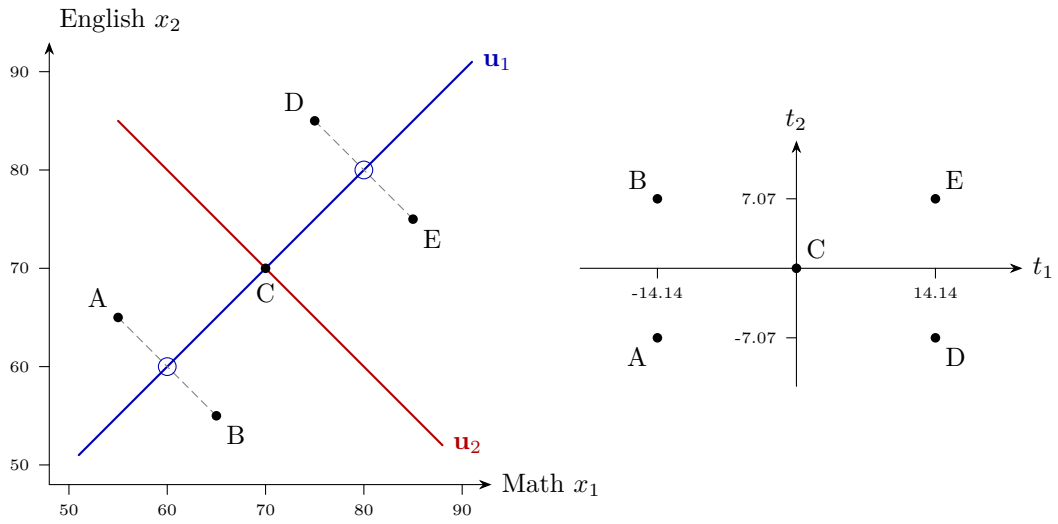
For illustration, take the mathematics and English scores of five students.

Student	Math x_1	English x_2	PC1 score	PC2 score
A	55	65	-14.14	-7.07
B	65	55	-14.14	7.07
C	70	70	0	0
D	75	85	14.14	-7.07
E	85	75	14.14	7.07

The sample mean and covariance are $\bar{\mathbf{x}} = (70, 70)^\top$, $\mathbf{S} = \begin{pmatrix} 125 & 75 \\ 75 & 125 \end{pmatrix}$, with eigenvalues and unit eigenvectors $\lambda_1 = 200$, $\mathbf{u}_1 = \frac{1}{\sqrt{2}}(1,1)^\top$, $\lambda_2 = 50$, $\mathbf{u}_2 = \frac{1}{\sqrt{2}}(1,-1)^\top$. The scores of each student are

$$t_{i1} = \frac{(x_{i1} - 70) + (x_{i2} - 70)}{\sqrt{2}}, \qquad t_{i2} = \frac{(x_{i1} - 70) - (x_{i2} - 70)}{\sqrt{2}}.$$

The first principal component expresses the **overall level** of the two scores, the second the **relative difference** between mathematics and English. Students A and B have the same first score but different second scores: with the first principal component alone, the difference in their relative strengths cannot be distinguished.



(a) original scores (b) principal component scores

Figure. *(a) The five students with the principal axes through $\bar{\mathbf{x}} = (70, 70)$; dashed segments are the orthogonal projections onto the first axis, and A, B (and D, E) project to the same point. (b) The score plot (t_{i1}, t_{i2}): PC1 (80% of the variance) is the overall level, PC2 (20%) the math–English difference that separates A from B.*

6.3 Dimension reduction and reconstruction error

For student E, $\mathbf{x}_E = (85, 75)^\top$ and the first principal component score is $t_{E1} = 10\sqrt{2}$. Reconstructing with the first principal component only,

$$\hat{\mathbf{x}}_E = \begin{pmatrix} 70 \\ 70 \end{pmatrix} + 10\sqrt{2} \cdot \frac{1}{\sqrt{2}} \begin{pmatrix} 1 \\ 1 \end{pmatrix} = \begin{pmatrix} 80 \\ 80 \end{pmatrix} :$$

the average level of the two scores is kept, but the difference between the subjects disappears.

Theorem 1.29: reconstruction error

For the reconstruction $\hat{\mathbf{x}}_i = \bar{\mathbf{x}} + \sum_{j \le q} t_{ij}\mathbf{u}_j$ from the first q principal components, the sum of squared reconstruction errors is

$$\sum_{i=1}^{n} \|\mathbf{x}_i - \hat{\mathbf{x}}_i\|^2 = (n-1) \sum_{j=q+1}^{p} \lambda_j.$$

Among all q-dimensional planes through the mean, PCA chooses the one that minimizes the reconstruction error; maximizing the total retained variance and minimizing the reconstruction error select the same subspace.

[Supplement]. $\mathbf{x}_i - \hat{\mathbf{x}}_i = \sum_{j>q} t_{ij}\mathbf{u}_j$, so $\|\mathbf{x}_i - \hat{\mathbf{x}}_i\|^2 = \sum_{j>q} t_{ij}^2$, and $\sum_i t_{ij}^2 = (n-1)\lambda_j$. For any $\mathbf{V} \in \mathbb{R}^{p\times q}$ with orthonormal columns, Pythagoras gives $\sum_i \|(\mathbf{x}_i - \bar{\mathbf{x}}) - \mathbf{V}\mathbf{V}^\top(\mathbf{x}_i - \bar{\mathbf{x}})\|^2 = (n-1)\{\mathrm{tr}\mathbf{S} - \mathrm{tr}(\mathbf{V}^\top\mathbf{S}\mathbf{V})\}$, so minimizing the error is maximizing $\mathrm{tr}(\mathbf{V}^\top\mathbf{S}\mathbf{V})$, whose maximum $\lambda_1 + \cdots + \lambda_q$ follows from von Neumann's inequality (Ch. 2) or Ch. 7. ■

6.4 Proportion of variance, scale and caveats

- The total sample variance is $\mathrm{tr}(\mathbf{S}) = \sum_{k=1}^{p} \lambda_k$, so the **proportion of variance explained** by the jth principal component is $\lambda_j / \sum_{k=1}^{p} \lambda_k$ and the **cumulative proportion** of the first q components is $\sum_{j=1}^{q} \lambda_j / \sum_{k=1}^{p} \lambda_k$. In the test-score example the first component explains $200/(200+50) = 80\%$. Choose q where the cumulative proportion reaches a chosen level, or from the decline in a **scree plot** of the ordered eigenvalues. For visualization one usually takes $q = 2$ or 3 and checks how much of the total variation these components explain.

- PCA based on the covariance matrix depends on the **units and scales** of the variables: if the English test is scored out of 1000 instead of 100, its variance is multiplied by 100 and the principal directions change. One can instead run PCA on the standardized data $z_{ik} = (x_{ik} - \bar{x}_k)/s_k$, $k = 1, \ldots, p$, $s_k > 0$, whose covariance matrix is the sample correlation matrix of the original data (**correlation PCA**). Standardizing is useful when units are arbitrary, but when differences in variance are themselves important information it may be inappropriate.

- **Does a high explained variance mean good group separation?** Let $Y \in \{-1, 1\}$ with $P(Y = 1) = P(Y = -1) = 1/2$ label two groups, and let $X_1 = 10Z$, $X_2 = Y$, $Z \sim N(0,1)$, $Z \perp\!\!\!\perp Y$. The population covariance and the proportion explained by the first component are $\mathbf{\Sigma} = \begin{pmatrix} 100 & 0 \\ 0 & 1 \end{pmatrix}$, $\frac{\lambda_1}{\lambda_1+\lambda_2} = \frac{100}{101} \approx 99.0\%$. The first component X_1 explains almost all of the variation but is independent of Y, while the second component X_2 alone separates the groups perfectly. PCA looks for directions that preserve much of the variation; a high proportion of variance does not guarantee that the information needed for group separation or prediction is kept.

Example 1.30: PCA of the gene expression data (slides Example 1.8)

Apply PCA to the Golub gene expression data (slides Example 1.2; $p = 7129$, $n = 72$) and draw the scatterplot of the first two principal component scores. Compared with the scatterplot of the first two original variables, the score plot separates ALL and AML much more clearly. PCA does not use the category labels, so this example shows that the principal components can be used to *explore* group differences.

7 Random Vectors and the Geometry of Variable Vectors

Motivation. Viewing each variable as a vector in $\mathbb{R}^n$ gives covariance and correlation a geometric meaning: lengths are standard deviations and angles are correlations. This picture makes projection arguments (regression, partial correlation) transparent.

7.1 Random vectors

In multivariate statistics the data $\mathbf{X} = [\mathbf{x}_1, \ldots, \mathbf{x}_n]^\top$ are regarded as a random sample from a population. The population from which multivariate data on p variables are drawn is modelled by the joint distribution of p random variables $X_1, \ldots, X_p$: collecting them in the random vector $\mathbf{x} = (X_1, \ldots, X_p)^\top$, its joint probability density function $f : \mathbb{R}^p \to [0, \infty)$ determines the distribution, written $\mathbf{x} = (X_1, \ldots, X_p)^\top \sim f$.

Example 1.31: distributions of random vectors (slides Example 2.1)

1. If $X_1, \ldots, X_p$ are independent with $X_i \sim \text{Unif}(a_i, b_i)$, the joint density is $f(x_1, \ldots, x_p) = \prod_{i=1}^p \frac{1}{b_i - a_i} \mathbf{1}\{x_i \in (a_i, b_i)\}$, where $b_i > a_i$ and $\mathbf{1}_{x \in (a,b)}$ is the indicator function: $\mathbf{x}$ is uniform on the hypercube $(a_1, b_1) \times \cdots \times (a_p, b_p)$.

2. If $X_1, \ldots, X_p$ are independent with $X_i \sim N(0, 1)$, the joint density is $f(x_1, \ldots, x_p) = \prod_i \frac{1}{\sqrt{2\pi}} e^{-x_i^2/2} = (2\pi)^{-p/2} e^{-(x_1^2 + \cdots + x_p^2)/2}$, the **standard multivariate normal distribution**.

3. If $(X_1, X_2)^\top$ is uniform on the triangle $0 < x_1 < x_2 < 1$, the joint density is $f(x_1, x_2) = 2 \cdot \mathbf{1}\{0 < x_1 < x_2 < 1\}$ (the triangle has area $1/2$, so the density is 2). Then $X_1 \mid X_2 = x_2 \sim \text{Unif}(0, x_2)$, $0 < x_2 < 1$: the conditional distribution of X_1 depends on the value of X_2, so the two components are not independent.

4. An example of a bivariate normal distribution with correlation ρ, $0 < |\rho| < 1$, has density $f(x_1, x_2) = \frac{1}{2\pi\sqrt{1-\rho^2}} \exp\{-\frac{x_1^2 - 2\rho x_1 x_2 + x_2^2}{2(1-\rho^2)}\}$, $(x_1, x_2) \in \mathbb{R}^2$. Each marginal is $X_1 \sim N(0, 1)$, $X_2 \sim N(0, 1)$, but the joint density does not factor into the product of the marginals, so the two components are not independent.

If n random vectors $\mathbf{x}_1, \ldots, \mathbf{x}_n$ are a simple random sample from a population with density f, they are mutually independent and identically distributed (**i.i.d.**), written $\mathbf{x}_1, \ldots, \mathbf{x}_n \overset{\text{iid}}{\sim} f$.

Definition 1.32: moments of a random vector

For a random vector $\mathbf{X} = (X_1, \ldots, X_p)^\top$ (and a random vector $\mathbf{Y}$):

$$\boldsymbol{\mu} = \mathrm{E}(\mathbf{X}) = \begin{pmatrix} \mathrm{E}(X_1) \\ \vdots \\ \mathrm{E}(X_p) \end{pmatrix}, \qquad \mathrm{Cov}(\mathbf{X}) = \mathrm{E}\{(\mathbf{X} - \boldsymbol{\mu})(\mathbf{X} - \boldsymbol{\mu})^\top\} = \mathrm{E}(\mathbf{X}\mathbf{X}^\top) - \mathrm{E}(\mathbf{X})\mathrm{E}(\mathbf{X})^\top,$$

$$\mathrm{Cov}(\mathbf{X}, \mathbf{Y}) = \mathrm{E}\{(\mathbf{X} - \mathrm{E}\mathbf{X})(\mathbf{Y} - \mathrm{E}\mathbf{Y})^\top\} = \mathrm{Cov}(\mathbf{Y}, \mathbf{X})^\top.$$

For $\boldsymbol{\Sigma} = \mathrm{Cov}(\mathbf{X}) \succ 0$ the Mahalanobis distance of $\mathbf{x}$ from $\boldsymbol{\mu}$ is $d_{\boldsymbol{\Sigma}}(\mathbf{x}, \boldsymbol{\mu}) = \sqrt{(\mathbf{x} - \boldsymbol{\mu})^\top \boldsymbol{\Sigma}^{-1} (\mathbf{x} - \boldsymbol{\mu})}$. The **generalized variance** is $|\mathrm{Cov}(\mathbf{X})|$ and the **total variance** is $\mathrm{tr}\{\mathrm{Cov}(\mathbf{X})\}$; their sample versions are $|\mathbf{S}|$ and $\mathrm{tr}(\mathbf{S}) = \sum_j \lambda_j$.

Theorem 1.33: characterization of covariance matrices

$\boldsymbol{\Sigma}$ is the covariance matrix of some random vector if and only if $\boldsymbol{\Sigma}$ is symmetric and NND.

Proof. ($\Rightarrow$) [Supplement] $\mathrm{Cov}(\mathbf{X})^\top = \mathrm{Cov}(\mathbf{X})$, and for every $\mathbf{a}$, $\mathbf{a}^\top\boldsymbol{\Sigma}\mathbf{a} = \mathrm{E}\{\mathbf{a}^\top(\mathbf{X}-\boldsymbol{\mu})\}^2 = \mathrm{Var}(\mathbf{a}^\top\mathbf{X}) \geq 0$.

($\Leftarrow$) As in the Remark on the existence of a Gaussian with covariance $\mathbf{S}$: $\mathbf{X} = \boldsymbol{\Sigma}^{1/2}\mathbf{Z}$ with $\mathbf{Z} \sim N_p(\mathbf{0}, \mathbf{I}_p)$ has $\mathrm{Cov}(\mathbf{X}) = \boldsymbol{\Sigma}$. ■

7.2 Mean, variance and correlation through variable vectors

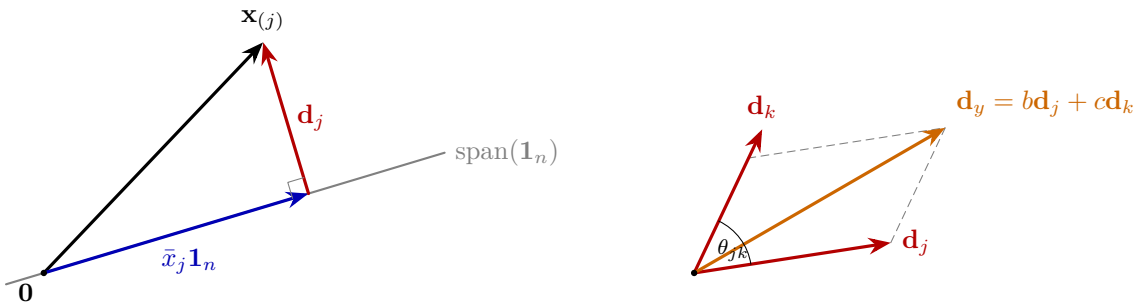
Using the variable vectors $\mathbf{x}_{(j)} \in \mathbb{R}^n$, that is, viewing the data as p vectors in n-dimensional space, gives a geometric reading of the statistics.

- **Mean**: the sample mean of variable j is $\bar{x}_j = \frac{1}{n}\sum_{i=1}^n x_{ij} = \frac{1}{n}\mathbf{1}_n^\top\mathbf{x}_{(j)} = (\mathbf{1}_n^\top\mathbf{1}_n)^{-1}\mathbf{1}_n^\top\mathbf{x}_{(j)}$, the coefficient of the orthogonal projection of $\mathbf{x}_{(j)}$ onto $\mathbf{1}_n$: $P_{\mathbf{1}_n}(\mathbf{x}_{(j)}) = \mathbf{1}_n(\mathbf{1}_n^\top\mathbf{1}_n)^{-1}\mathbf{1}_n^\top\mathbf{x}_{(j)} = \bar{x}_j\mathbf{1}_n$. The projection score of $\mathbf{x}_{(j)}$ onto $\mathrm{span}(\mathbf{1}_n)$ is $s_{\mathbf{1}}(\mathbf{x}_{(j)}) = \sqrt{n}\,\bar{x}_j$.
- **Deviation vector**: the deviation of $\mathbf{x}_{(j)}$ from its "mean vector" $\bar{x}_j\mathbf{1}_n$ is $\mathbf{d}_j = \mathbf{x}_{(j)} - \bar{x}_j\mathbf{1}_n = (x_{1j}-\bar{x}_j, \ldots, x_{nj}-\bar{x}_j)^\top$, and $\mathbf{x}_{(j)} = \bar{x}_j\mathbf{1}_n + \mathbf{d}_j$ is an orthogonal decomposition ($\mathbf{1}_n^\top\mathbf{d}_j = 0$). The deviation vectors carry the variability of the variables (standard deviation, variance) and the linear relations between pairs of variables (covariance, correlation).
- The sample standard deviation is proportional to the length of the deviation vector, $s_j = \frac{1}{\sqrt{n-1}}\|\mathbf{d}_j\|$; the sample variance is proportional to its squared length, $s_j^2 = \frac{1}{n-1}\|\mathbf{d}_j\|^2$; and the sample covariance is proportional to the inner product of two deviation vectors, $s_{jk} = \frac{1}{n-1}\mathbf{d}_j^\top\mathbf{d}_k = \frac{1}{n-1}\langle\mathbf{d}_j, \mathbf{d}_k\rangle$.
- **The sample correlation is the cosine of the angle between deviation vectors**: with the angle θ_{jk} between $\mathbf{d}_j$ and $\mathbf{d}_k$, $\mathbf{d}_j^\top\mathbf{d}_k = \|\mathbf{d}_j\|\|\mathbf{d}_k\|\cos\theta_{jk}$, so

$$r_{jk} = \frac{s_{jk}}{s_j s_k} = \frac{\mathbf{d}_j^\top\mathbf{d}_k}{\|\mathbf{d}_j\|\|\mathbf{d}_k\|} = \cos\theta_{jk}.$$

- The raw cross product splits accordingly: $\mathbf{x}_{(j)}^\top\mathbf{x}_{(k)} = n\bar{x}_j\bar{x}_k + \mathbf{d}_j^\top\mathbf{d}_k$, i.e. $\frac{n-1}{n}\,s_{jk} = \overline{x_jx_k} - \bar{x}_j\bar{x}_k$ with $\overline{x_jx_k} = \frac{1}{n}\sum_i x_{ij}x_{ik}$.

Geometric meaning: a longer deviation vector $\mathbf{d}_j$ means that the values of variable j lie farther from their mean, i.e. more variability. If $\mathbf{d}_j$ and $\mathbf{d}_k$ are parallel, $\mathbf{d}_j = c\mathbf{d}_k$ for some $c \neq 0$ and $\theta = 0$ (or π), and x_j, x_k are perfectly linearly related: $r_{jk} = +1$ for $\theta = 0$ (same direction) and $r_{jk} = -1$ for $\theta = \pi$ (opposite directions). Conversely, if x_j, x_k are perfectly linearly related, their deviation vectors are parallel. If $\mathbf{d}_j^\top\mathbf{d}_k = 0$, the deviation vectors are orthogonal and the *linear* correlation is $r_{jk} = 0$.



(a) $\mathbf{x}_{(j)} = \bar{x}_j\mathbf{1}_n + \mathbf{d}_j$ (b) $r_{jk} = \cos\theta_{jk}$, $s_j \propto \|\mathbf{d}_j\|$

Figure. *Variables as vectors in $\mathbb{R}^n$. (a) Projecting a variable vector on $\mathbf{1}_n$ gives its mean part; the orthogonal remainder is the deviation vector. (b) Lengths of deviation vectors are proportional to standard deviations, the cosine of the angle between them is the sample correlation, and a linear combination $y = a + bx_j + cx_k$ has deviation vector $b\mathbf{d}_j + c\mathbf{d}_k$.*

Theorem 1.34: unbiased estimation of the covariance matrix (slides Result 2.1)

Let $\mathbf{X}_1, \ldots, \mathbf{X}_n$ be a random sample from a joint distribution with mean vector $\boldsymbol{\mu}$ and covariance matrix $\boldsymbol{\Sigma}$. Then $\bar{\mathbf{X}}$ is an unbiased estimator of $\boldsymbol{\mu}$,

$$\mathrm{E}(\bar{\mathbf{X}}) = \boldsymbol{\mu}, \qquad \mathrm{Cov}(\bar{\mathbf{X}}) = \tfrac{1}{n}\boldsymbol{\Sigma},$$

(the population covariance divided by the sample size), and $\mathbf{S}_n$ is a biased estimator:

$$\mathrm{E}(\mathbf{S}_n) = \tfrac{n-1}{n}\boldsymbol{\Sigma} = \boldsymbol{\Sigma} - \tfrac{1}{n}\boldsymbol{\Sigma}, \qquad \mathrm{E}\Big(\tfrac{n}{n-1}\mathbf{S}_n\Big) = \mathrm{E}(\mathbf{S}) = \boldsymbol{\Sigma}.$$

Proof. Recall $\bar{\mathbf{X}} = (\mathbf{X}_1 + \mathbf{X}_2 + \cdots + \mathbf{X}_n)/n$. Repeated use of the properties of expectation gives $\mathrm{E}(\bar{\mathbf{X}}) = \mathrm{E}(\frac{1}{n}\mathbf{X}_1 + \cdots + \frac{1}{n}\mathbf{X}_n) = \frac{1}{n}\mathrm{E}(\mathbf{X}_1) + \cdots + \frac{1}{n}\mathrm{E}(\mathbf{X}_n) = \frac{1}{n}\boldsymbol{\mu} + \cdots + \frac{1}{n}\boldsymbol{\mu} = \boldsymbol{\mu}$. Next,

$$(\bar{\mathbf{X}} - \boldsymbol{\mu})(\bar{\mathbf{X}} - \boldsymbol{\mu})^\top = \Big(\frac{1}{n}\sum_{j=1}^{n}(\mathbf{X}_j - \boldsymbol{\mu})\Big)\Big(\frac{1}{n}\sum_{\ell=1}^{n}(\mathbf{X}_\ell - \boldsymbol{\mu})\Big)^\top = \frac{1}{n^2}\sum_{j=1}^{n}\sum_{\ell=1}^{n}(\mathbf{X}_j - \boldsymbol{\mu})(\mathbf{X}_\ell - \boldsymbol{\mu})^\top,$$

so $\mathrm{Cov}(\bar{\mathbf{X}}) = \mathrm{E}\{(\bar{\mathbf{X}} - \boldsymbol{\mu})(\bar{\mathbf{X}} - \boldsymbol{\mu})^\top\} = \frac{1}{n^2}\sum_j\sum_\ell \mathrm{E}\{(\mathbf{X}_j - \boldsymbol{\mu})(\mathbf{X}_\ell - \boldsymbol{\mu})^\top\}$. For $j \neq \ell$, $\mathrm{E}\{(\mathbf{X}_j - \boldsymbol{\mu})(\mathbf{X}_\ell - \boldsymbol{\mu})^\top\} = \mathbf{0}$ by the independence of $\mathbf{X}_j$ and $\mathbf{X}_\ell$. Therefore $\mathrm{Cov}(\bar{\mathbf{X}}) = \frac{1}{n^2}(n\boldsymbol{\Sigma}) = \frac{1}{n}\boldsymbol{\Sigma}$, where $\boldsymbol{\Sigma}$ is the common population covariance matrix.

[Supplement] For $\mathbf{S}_n$: $\sum_j(\mathbf{X}_j - \bar{\mathbf{X}})(\mathbf{X}_j - \bar{\mathbf{X}})^\top = \sum_j(\mathbf{X}_j - \boldsymbol{\mu})(\mathbf{X}_j - \boldsymbol{\mu})^\top - n(\bar{\mathbf{X}} - \boldsymbol{\mu})(\bar{\mathbf{X}} - \boldsymbol{\mu})^\top$ (expand $\mathbf{X}_j - \bar{\mathbf{X}} = (\mathbf{X}_j - \boldsymbol{\mu}) - (\bar{\mathbf{X}} - \boldsymbol{\mu})$ and use $\sum_j(\mathbf{X}_j - \boldsymbol{\mu}) = n(\bar{\mathbf{X}} - \boldsymbol{\mu})$). Its expectation is $n\boldsymbol{\Sigma} - n \cdot \frac{1}{n}\boldsymbol{\Sigma} = (n-1)\boldsymbol{\Sigma}$, so $\mathrm{E}(\mathbf{S}_n) = \frac{n-1}{n}\boldsymbol{\Sigma}$ and $\mathrm{E}(\mathbf{S}) = \boldsymbol{\Sigma}$. ■

7.3 Linear combinations of variables

When a new variable y is formed as a linear combination of variables, its mean, variance and covariance with other variables can be understood geometrically. For example, let the ith observation of a new variable be the linear combination of the first two variables

$$y_i = a + bx_{i1} + cx_{i2}.$$

The variable vector of y is the corresponding combination of variable vectors, $\mathbf{y} = (y_1, \ldots, y_n)^\top = a\mathbf{1}_n + b\mathbf{x}_{(1)} + c\mathbf{x}_{(2)}$. Recalling $\mathbf{x}_{(1)} = P_{\mathbf{1}_n}(\mathbf{x}_{(1)}) + \mathbf{d}_1$, its mean vector is

$$P_{\mathbf{1}_n}(\mathbf{y}) = a\mathbf{1}_n + bP_{\mathbf{1}_n}(\mathbf{x}_{(1)}) + cP_{\mathbf{1}_n}(\mathbf{x}_{(2)}) = (a + b\bar{x}_1 + c\bar{x}_2)\mathbf{1}_n,$$

and its deviation vector is $\mathbf{d}_y = b\mathbf{d}_1 + c\mathbf{d}_2$. The sample standard deviation of y is proportional to the length of $\mathbf{d}_y$, and with $\mathbf{d}_y^\top \mathbf{d}_y = (b\mathbf{d}_1 + c\mathbf{d}_2)^\top (b\mathbf{d}_1 + c\mathbf{d}_2)$ it can be written through the standard deviations and correlation of x_1, x_2:

$$\widehat{\mathrm{Var}}(y) = \frac{1}{n-1}\mathbf{d}_y^\top \mathbf{d}_y = \frac{b^2\|\mathbf{d}_1\|^2 + c^2\|\mathbf{d}_2\|^2 + 2bc\|\mathbf{d}_1\|\|\mathbf{d}_2\|\cos\theta}{n-1}.$$

$\sqrt{\widehat{\mathrm{Var}}(y)}$ is the sample standard deviation of the data $y_1, \ldots, y_n$. (The slides sketch $\mathbf{d}_y$ for $y = 2x_1 + x_2$ in the space of individuals.)

More generally, for constants a and $\mathbf{b} \in \mathbb{R}^p$ the new variable $z = a + \mathbf{b}^\top \mathbf{x} = a + \sum_{j=1}^p b_j x_j$ has variable vector $\mathbf{z} = (z_1, \ldots, z_n)^\top = a\mathbf{1}_n + \sum_j b_j \mathbf{x}_{(j)}$, with

$$P_{\mathbf{1}_n}(\mathbf{z}) = \Big(a + \sum_{j=1}^p b_j \bar{x}_j\Big)\mathbf{1}_n, \qquad \mathbf{d}_z = \sum_{j=1}^p b_j \mathbf{d}_j = [\mathbf{d}_1, \ldots, \mathbf{d}_p]\begin{pmatrix} b_1 \\ \vdots \\ b_p \end{pmatrix} = \mathbf{X}_C\mathbf{b}.$$

Theorem 1.35: sample variance and covariance of linear combinations

$$\widehat{\mathrm{Var}}(z) = \tfrac{1}{n-1}\mathbf{d}_z^\top \mathbf{d}_z = \tfrac{1}{n-1}\mathbf{b}^\top \mathbf{X}_C^\top \mathbf{X}_C \mathbf{b} = \mathbf{b}^\top \mathbf{S}\mathbf{b};$$

for two new variables $y = \mathbf{a}^\top \mathbf{x}$ and $z = \mathbf{b}^\top \mathbf{x}$ $(\mathbf{a}, \mathbf{b} \in \mathbb{R}^p)$, $\widehat{\mathrm{Cov}}(y, z) = \tfrac{1}{n-1}\mathbf{d}_y^\top \mathbf{d}_z = \tfrac{1}{n-1}\mathbf{a}^\top \mathbf{X}_C^\top \mathbf{X}_C \mathbf{b} = \mathbf{a}^\top \mathbf{S}\mathbf{b}$.

8 Dimension and Rank of the Data

Motivation. If $n \le p$, the sample covariance matrix is singular and cannot be inverted. The rank result here explains when classical multivariate procedures break down.

- **Linear independence**: a set of vectors $\{\mathbf{x}_1, \ldots, \mathbf{x}_m\}$ is linearly independent if no vector $\mathbf{x}_i$ is a linear combination of the others, i.e. $\sum_{i=1}^m c_i \mathbf{x}_i = \mathbf{0}$ implies $c_i = 0$ for all i.
- **Rank**: the rank of a matrix $\mathbf{X}$ is the maximal number of linearly independent columns (or rows). Column rank equals row rank for every matrix, and $\mathrm{rank}(\mathbf{X}) \le \min(n, p)$.
- The centered data matrix $\mathbf{X}_C = [\mathbf{x}_1 - \bar{\mathbf{x}}, \ldots, \mathbf{x}_n - \bar{\mathbf{x}}]^\top$ removes the mean in every dimension, adjusting the data to center $\mathbf{0}$. Its columns $\mathbf{d}_1, \ldots, \mathbf{d}_p$ lie in the subspace of $\mathbb{R}^n$ orthogonal to $\mathbf{1}_n$, whose span determines the rank: $\mathrm{rank}(\mathbf{X}_C) = \dim \mathrm{span}(\mathbf{d}_1, \ldots, \mathbf{d}_p) \le \min(n-1, p)$.
- $\mathrm{rank}(\mathbf{A}) = \mathrm{rank}(\mathbf{A}^\top) = \mathrm{rank}(\mathbf{A}^\top\mathbf{A})$. Hence $\mathrm{rank}(\mathbf{X}_C)$ is both its row and column rank, and $\mathrm{rank}(\mathbf{S}) = \mathrm{rank}(\mathbf{X}_C^\top \mathbf{X}_C) = \mathrm{rank}(\mathbf{X}_C)$.
- An $m \times k$ matrix $\mathbf{A}$ is **full-rank** if $\mathrm{rank}(\mathbf{A}) = \min(m, k)$ and **rank-deficient** if $\mathrm{rank}(\mathbf{A}) < \min(m, k)$.

If the sample covariance $\mathbf{S}$ is rank-deficient, $\mathrm{rank}(\mathbf{S}) = r$ is smaller than p: some variables are perfectly linearly related, so the variation of the data lies in an r-dimensional subspace of the p-dimensional data space and is zero in the remaining $p - r$ dimensions. (The slides show a 3-dimensional input cloud lying on a curved 2-dimensional sheet, mapped to a 2-dimensional embedded space.)

Theorem 1.36: rank-deficiency property (slides Proposition, homework)

For the NND sample covariance $\mathbf{S}$ of $\mathbf{X} = [\mathbf{x}_1, \dots, \mathbf{x}_n]^\top \in \mathbb{R}^{n\times p}$ the following are equivalent.

1. $\mathbf{S}$ is rank-deficient.
2. There is $\mathbf{a} \in \mathbb{R}^p \setminus \{\mathbf{0}\}$ with $\mathbf{a}^\top(\mathbf{x}_i - \bar{\mathbf{x}}) = 0$ for all $i = 1, \dots, n$.
3. The projection scores onto some direction all coincide: $s_{\mathbf{a}}(\mathbf{x}_i) = s_{\mathbf{a}}(\mathbf{x}_j)$ for all $i, j = 1, \dots, n$.
4. The subspace $\operatorname{span}(\mathbf{d}_1, \dots, \mathbf{d}_p)$ spanned by the deviation vectors has dimension less than p.
5. At least one deviation vector $\mathbf{d}_j$ is a linear combination of the others.
6. At least one eigenvalue of $\mathbf{S}$ is 0.

Proof. (1)⇔(2): If $\mathbf{S}$ is rank-deficient, $\operatorname{rank}(\mathbf{S}) = \operatorname{rank}(\mathbf{X}_C^\top) = r < p$, so the column vectors $\{\mathbf{x}_i - \bar{\mathbf{x}} : i = 1, \dots, n\}$ of $\mathbf{X}_C^\top$ span a subspace of dimension r. Hence there is a vector $\mathbf{a} \in \mathbb{R}^p \setminus \{\mathbf{0}\}$ orthogonal to that subspace, so $\mathbf{a}^\top(\mathbf{x}_i - \bar{\mathbf{x}}) = 0$ for all i. Conversely, if such an $\mathbf{a}$ exists, the vectors $\{\mathbf{x}_i - \bar{\mathbf{x}}\}$ lie in a subspace of dimension less than p, and $\mathbf{S}$ is rank-deficient.

(2)⇔(3): $\mathbf{a}^\top(\mathbf{x}_i - \bar{\mathbf{x}}) = 0$ for all i iff $\mathbf{a}^\top\mathbf{x}_i = \mathbf{a}^\top\bar{\mathbf{x}}$ for all i, iff all scores $s_{\mathbf{a}}(\mathbf{x}_i) = \mathbf{a}^\top\mathbf{x}_i/\|\mathbf{a}\|$ coincide (their common value is then the mean score).

(1)⇔(4): $\operatorname{rank}(\mathbf{S}) = \operatorname{rank}(\mathbf{X}_C) = \operatorname{rank}([\mathbf{d}_1, \dots, \mathbf{d}_p])$, so by the definition of rank (1)⇔(4).

(4)⇔(5): $\operatorname{span}(\mathbf{d}_1, \dots, \mathbf{d}_p)$ has dimension less than p iff $\{\mathbf{d}_1, \dots, \mathbf{d}_p\}$ is linearly dependent, so (4)⇔(5) by the definition of linear independence.

(2)⇒(6): For $\mathbf{a}$ as in (2), $\mathbf{X}_C\mathbf{a} = \mathbf{0}$, so $\mathbf{S}\mathbf{a} = \frac{1}{n-1}\mathbf{X}_C^\top\mathbf{X}_C\mathbf{a} = \mathbf{0} = 0 \cdot \mathbf{a}$: $\mathbf{a}$ is an eigenvector of $\mathbf{S}$ with eigenvalue 0.

(6)⇒(2): If an eigenvalue of $\mathbf{S}$ is 0 with eigenvector $\mathbf{u}$, then $\mathbf{S}\mathbf{u} = \mathbf{0}$, and $\mathbf{u}^\top\mathbf{S}\mathbf{u} = \frac{1}{n-1}\|\mathbf{X}_C\mathbf{u}\|^2 = 0$. The vector $\mathbf{X}_C\mathbf{u}$ has length 0, so all its components $\mathbf{u}^\top(\mathbf{x}_i - \bar{\mathbf{x}})$ are 0: $\mathbf{u}$ is a vector as in (2). ■

Statistical meaning. If $\mathbf{S}$ is full rank, the data have enough independent variation in every direction, and every component of the data contributes meaningfully to the analysis. If $\mathbf{S}$ is rank-deficient, the data give no information in some direction and the remaining directions are strongly correlated: compared with their size, the data provide insufficient information, there is little independent variation, and the reliability of statistical analysis drops. In particular $\mathbf{S}^{-1}$ does not exist, so the sphering $\mathbf{x} \mapsto \mathbf{S}^{-1/2}\mathbf{x}$ is undefined. In high dimensions ($p > n$) $\mathbf{S}$ is always rank-deficient:

$$\operatorname{rank}(\mathbf{S}) \le \min\{n - 1, p\} < p.$$

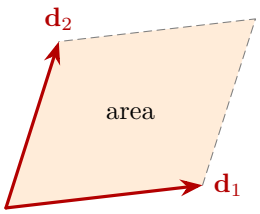
Remark 1.37: generalized variance as a squared volume [Supplement]

For $p = 2$, $|\mathbf{S}| = s_{11}s_{22} - s_{12}^2 = \frac{\|\mathbf{d}_1\|^2\|\mathbf{d}_2\|^2 - (\mathbf{d}_1^\top\mathbf{d}_2)^2}{(n-1)^2} = \frac{(\|\mathbf{d}_1\|\|\mathbf{d}_2\|\sin\theta_{12})^2}{(n-1)^2}$: the generalized variance is the squared area of the parallelogram spanned by the deviation vectors, divided by $(n-1)^2$. In general $|\mathbf{S}| = (n-1)^{-p}(\text{volume})^2$, where the volume is that of the parallelotope spanned by $\mathbf{d}_1, \dots, \mathbf{d}_p$.

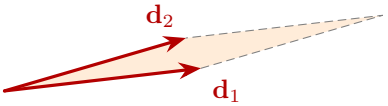
Proof for general p. Write $\mathbf{X}_C = [\mathbf{d}_1, \dots, \mathbf{d}_p] = \mathbf{Q}\mathbf{R}_0$ with $\mathbf{Q} \in \mathbb{R}^{n\times p}$ having orthonormal columns and $\mathbf{R}_0$ upper triangular (Gram–Schmidt; if the $\mathbf{d}_j$ are dependent, some diagonal entry of $\mathbf{R}_0$ is 0 and both sides below vanish). Then $|\mathbf{S}| = (n-1)^{-p}|\mathbf{X}_C^\top\mathbf{X}_C| = (n-1)^{-p}|\mathbf{R}_0^\top\mathbf{R}_0| = (n-1)^{-p}\prod_j(\mathbf{R}_0)_{jj}^2$, and $|(\mathbf{R}_0)_{jj}|$ is the distance from $\mathbf{d}_j$ to $\operatorname{span}(\mathbf{d}_1, \dots, \mathbf{d}_{j-1})$, the jth "height". The product of the heights is the volume.

So $|\mathbf{S}|$ is small when some deviation vector is nearly a combination of the others (strong

correlation), even if every variance is large, and $|\mathbf{S}| = 0$ exactly in the rank-deficient case of Theorem 1.36. Two data sets can share $\mathrm{tr}\mathbf{S}$ while having very different $|\mathbf{S}|$.



weak correlation: $|\mathbf{S}|$ large



strong correlation: $|\mathbf{S}|$ small

Figure. [Supplement] *For $p = 2$, $|\mathbf{S}| = (\text{area})^2/(n-1)^2$ with the parallelogram spanned by the deviation vectors. With equal lengths (equal variances), a small angle between $\mathbf{d}_1$ and $\mathbf{d}_2$ (correlation near ± 1) flattens the parallelogram and the generalized variance tends to 0.*